

CQUNIVERSITY AUSTRALIA

CQU ROCKHAMPTON TAFE CONSOLIDATION - STAGE 2

FIRE PROTECTION SERVICES PERFORMANCE SPECIFICATION

FEBRUARY 2026

CONFIDENTIAL



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CQU Rockhampton TAFE Consolidation - Stage 2 Fire Protection Services Performance Specification

CQUUniversity Australia

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1 PROJECT INFORMATION

1.1 GENERAL

WSP Australia Pty Limited WSP has been engaged for the CQU TAFE Redevelopment Project as part of design team to deliver these specifications considering existing and future demands. Based on a 2022 masterplan study, the scope includes the provision of a new Trades Building (Building 92), and the refurbishment of Buildings 19, 33, 34, 36 and Engineering & Horticulture.

As part of the initial scope of services, WSP attended the Rockhampton North and City Campuses to undertake a high-level review of the existing Fire Protection Services infrastructure reticulating around each site.

This specification takes into consideration existing items of plant and equipment, and its capacity for the proposed expansion.

The project delivery being design and construct, has been adopted for this project.

This Specification is to be read in conjunction with other tender documents issued by the Project Manager including:

- Principal's Project Requirements
- Conditions of Tender
- Conditions of Contract
- Contract Preliminaries

This Specification is a technical specification and for the purpose of this Specification, Works shall mean all works or work as described in this Specification and/or associated drawings

This technical specification and drawings are intended for a Design and Construct Contract and have been provided to communicate the design intent of the Works for the project and for the guidance of tenderers pricing the project.

Notwithstanding this, the completion of the design, the co-ordination of the electrical services works, and achieving the system performance covered by this technical specification & Drawings shall form part of the Works.

The drawings and scope of works are based on information and drawings provided by the Client, other stakeholders and other design consultants which may contain conflicting data, incomplete design, coordination, or omissions. These layouts will be subject to change to suit final design. The contractor pricing the works shall make an assessment and make adequate contingency based on their experience with the procurement and risks involved in projects of this nature i.e., changes to requirements, site investigation outcomes, scope, and future design detail.

The Works requires the necessary resources to complete the detailed "For Construction" documentation and schedules, final equipment selections, testing and commissioning of the installation in accordance with the design intent and the requirements of this technical specification and drawings.

Notwithstanding this, achieving the system performance covered by this technical specification & drawings shall form part of the Works.

The work shall comprise that which is described in this document, any subsequent addendums.

The Drawings shall be read in conjunction with the respective existing services drawings and manuals and all associated documentation.

The tenderer shall visit the site to fully familiarise themselves with existing services installation and site conditions. Allowance shall be made to coordinate the existing services with the project staging and the scope of works covered under this Contract.

Include for all applicable site allowances and overtime required to meet the project programme.

Tender prices are to be submitted as a lump sum, fixed for the duration of the contract.

1.1.1 ASSOCIATED DOCUMENTS

This technical specification is to be read in conjunction with the project documents and it shall be taken that for the purpose of this specification and scope of works all associated drawings and technical specifications including but not limited to those below have been received, perused, and acknowledged in the submission of a commercial offer for the scope of works:

- 1 Site Survey Drawings
- 2 Architectural and Interior Design
- 3 Structural & Civil Drawings
- 4 Mechanical Services
- 5 Fire Protection Services
- 6 Hydraulic Services
- 7 Fire Engineering Report

1.2 PROJECT TEAM

The Project Team includes the following team members:

PRINCIPAL/CLIENT	CQU
CLIENT REPRESENTATIVE	TBC
PROJECT MANAGER	TBC
CONTRACTOR	The company who undertakes the Works
ARCHITECT	Thompson Adsett & Peddle Thorp
SERVICES CONSULTANT ENGINEER (MECHANICAL, ELECTRICAL, FIRE PROTECTION, HYDRAULICS)	WSP Australia Pty Limited
CIVIL ENGINEER	WSP Australia Pty Limited
STRUCTURAL ENGINEER	WSP Australia Pty Limited
FIRE SAFETY ENGINEER	WSP Australia Pty Limited
BUILDING SURVEYOR	TBC

1.3 THE PROJECT DESCRIPTION

1.3.1 GENERAL DESCRIPTION

The project is the consolidation of the existing Rockhampton CQU and TAFE site. The project comprises of:

- A new Trades Centre Building
- Refurbishment Building 19 level 3
- Partial/small refurbishment of building 32
- Refurbishment Building 33 levels ground, 1 and 2
- Partial/small refurbishment of building 34 ground floor
- Refurbishment Building 36
- New Horticulture Compound

Project information

- Refurbishment of Engineering building
- Internal vehicle accessways & parking & associated landscape paths etc
- Associated services enclosures/facilities

1.3.1.1 UTILITIES

The following utilities are relevant for this project:

- Electricity – Existing HV network is privately owned by CQU
- Water
- Sewer
- Storm water
- Telecommunications
- Fire Authority

2 GENERAL REQUIREMENTS

2.1 GENERAL

2.1.1 DEFINITIONS

Unless the context otherwise requires, the following definitions apply:

- 1 Supply: 'Supply', 'furnish' and similar expressions mean 'supply only'.
- 2 Provide: 'Provide' and similar expressions mean 'supply, deliver and install' and include all testing, commissioning, tuning, interfaces etc.
- 3 Proprietary: 'Proprietary' means identifiable by naming manufacturer, supplier, installer, trade name, brand name, and catalogue or reference number.
- 4 Samples: Includes samples and prototypes.
- 5 Zinc-coated steel: Includes zinc-coated steel, zinc/iron alloy-coated steel and aluminium/zinc-coated steel.

2.1.2 SPECIFICATION OBJECTIVES

The intent of this Specification is to:

- 1 Provide a basis for competitive tendering for suitably qualified tenderers.
- 2 Allow competent contractors to price the works.
- 3 Provide a clear statement of technical and performance requirements against which compliance can be assessed.
- 4 Provide documentation required by Authorities to demonstrate design compliance with statutory requirements for the purpose of building permit application.
- 5 Provide documentation detailing the scope and quality of the project for the purpose of Client's confirmation of compliance to the Client's project brief.
- 6 Define the requirements of the Contractor with respect to Quality Assurance.
- 7 Define the scope of services including project management, installation engineering, testing commissioning, maintenance and documentation to be provided.
- 8 Specify the technical and performance requirements of the Installation.
- 9 The Specification relies upon the Contractor having the necessary specialist services resources to complete the detailed design, testing and commissioning of the installation in accordance with the design intent and the requirements of the specification.

2.1.3 DESIGN INTENT

Intent: The Contract documents are intended to call for complete, operational systems. Provide all items (major and minor), equipment, accessories and incidental work required for the completion of the installation and to ensure full integration.

Diagrammatic layouts: Layouts of service lines, plant and equipment shown on the drawings are diagrammatic only, except where figured dimensions are provided or calculable. Before commencing work, inspect the site, obtain all necessary measurements and information.

2.1.4 DISCREPANCIES IN DOCUMENTS

Give immediate notice in case of any ambiguity, discrepancy or inconsistency in the Contract document. Should conflict arise between the requirements of specification, drawings, mandatory requirements, standards including Standards Australia Codes, then the higher standard or most stringent requirements shall prevail and shall be deemed to have been included in the Contract. The Contract's order of documents precedence is not intended and shall not be applicable in this case

2.1.5 REFERENCED DOCUMENTS/STANDARDS

Use referenced documents/standards which are editions, with amendments, current 3 months before the closing date for tenders, except where other editions or amendments are required by Statutory Authorities.

2.2 QUALITY ASSURANCE

Provide a Quality Assurance plan in order to propose, establish, maintain, monitor and document a quality assurance system covering all aspects of the design, purchase, fabrication, installation and completion of the works. The plan shall be in accordance with ISO 9001/.2 (as appropriate) and comply with the Contract.

2.3 ELECTROMAGNETIC COMPATIBILITY

- 1 Comply with Australian Communications Authority requirements for electrical and electronics products to limit electromagnetic interference (EMI).
- 2 Emissions:
 - a Passive products with C-tick or Regulatory Compliance Mark (RCM) to AS/NZS 4417.3, marking of electrical products to indicate compliance with regulations – Specific requirements for electromagnetic compatibility regulatory applications.
- 3 Immunity:
 - a Electrical and electronic apparatus: To AS/NZS 4252.1, Electromagnetic compatibility – Generic immunity standard – Residential, commercial and light industry (EN 50082-1), or
 - b EN 5008-2, Electromagnetic compatibility – Generic immunity standard – Industrial environment.
- 4 Harmonics and Voltage Surges:
 - a Levels of emissions to be acceptable to the electricity distribution code. Keep the THDI current to 5% maximum for all individual components/equipment and at least to that level at the first supplying control panel for all components/equipment connected to the switchboard. Provide active harmonic filters within the component, or respective control panel. Do not rely on the external harmonic filters to be installed on the Main Switchboard(s).

2.4 BUILDING PENETRATIONS

2.4.1 FIRE RESISTING BUILDING ELEMENTS

- 1 Seal penetrations to comply with NCC using a system to AS 4072.1, Components for the protection of openings in fire-resistant separating elements – Services penetrations and control joints.
- 2 Comply with Section 3 of AS/NZS 1668:1, Fire and Smoke control in Multi Compartment Buildings.
- 3 AS4072.1 inter-alia Appendix B and AS1851 inter-alia Clause 17.2.3.1 shall be complied with in as-built documentation to identify all fire and smoke containment compartment walls and barriers. To satisfy specific

identification requirements to AS4072.1 Appendix B4 and AS1851 Clause 17.2.4, permanently fixed labelling, tagging and signage of all passive fire and smoke containment systems must be provided for on-site identification. Fire stopping barriers and penetrations shall comply in particular with reference to a service label affixed adjacent to each fire stop barrier and service penetration (or close proximity group) as detailed on page 25 of AS4072.1 and page 165 of AS1851. This as-built provision amongst other as-built documentation and manuals is essential as part of handing over process of the project.

2.4.2 *NON-FIRE RESISTING BUILDING ELEMENTS*

- 1 Seal penetrations around conduits and sleeves. Seal around cables within sleeves. If the building element is weatherproof, acoustic rated or subject to pressure, maintain the rating. Acoustically seal penetrations through plantroom walls and floors.

2.4.3 *CHASES*

- 1 Cut, drill and chase as necessary, where possible avoiding finished work. Where chasing of finished work is essential, make good all damage to the original standards of the finishes.

2.4.4 *LIMITATIONS*

- 1 General: Do not penetrate, or chase the following without approval:
- 2 Structural building elements including external walls, core walls, fire walls, floor slabs, beams or columns.
- 3 Acoustic barriers.
- 4 Other building services.
- 5 Membrane elements including damp-proof courses, waterproofing membranes and roof coverings.
- 6 Membranes: If approval is given to penetrate membranes, provide a waterproof seal between the membrane and the penetrating compound.

2.5 REQUIRED SUBMISSIONS

- 1 Provide in good time to allow review without impediment to the programme, including possible amendment and resubmission:-
 - a Copies of correspondence and notes of meetings with Authorities.
 - b Documents showing approval of the Authorities whose requirements may apply to the work.
 - c Certified schedule of compliance for all plant and equipment, prior to placing orders.
 - d Factory test results where applicable.
 - e All product data, performance test and commissioning results required by this Specification.
 - f Shop drawings for fabrication and installation of all equipment and items supplied.
 - g Manufacturer's product data for equipment, materials, components and systems including:
 - i Technical specifications and drawings.
 - ii Size, arrangement, operating and maintenance clearances.
 - iii Operating weight.
 - iv Type test reports.

General Requirements

- v Performance and rating curves or tables marked with each selection showing for both duty point and for part load conditions the capacity; the energy consumption and power factor; and the sound power level.
- vi Motor power requirements (both full load and starting).
- vii Control details.
- viii Recommendations for installation and maintenance.
- ix Evidence of compliance with specified product certification schemes.
- h Inspection, test and commissioning plans for every section of the works. Detail:
 - i The procedure of how to complete the task.
 - ii The skill or competency of the person undertaking the works.
 - iii The review or testing procedure to assure satisfactory completion of the task.
 - iv The person within the Fire Services Works Package organisation authorised to sign-off the task as accepted.
- i Client handover/training proposals.
- j Record and installation drawings: Record all changes to equipment and services layouts, wiring and any other items during the construction period, which may have been incorporated into these works.
- k Operating and Maintenance manuals.
- l Maintenance/service records during the defects liability period.
- 2 Give notice before commencing work affected by the submissions, unless the submissions have been reviewed with no exception taken.
- 3 Maintain and submit monthly a schedule of shop drawings and submissions to identify all proposed drawings and submissions designated in the following categories:
 - a Accepted.
 - b Submitted, awaiting acceptance (with submission date and revision number).
 - c Not yet submitted.
- 4 Submit electronically (PDF or alternative agreed format).
- 5 Provide identifications of the Contractor, supplier/manufacture, applicable product name, product number, included product options, and relevant specification references.
- 6 Where selected equipment or systems do not fully comply with this Specification and associated documents, submit details of proposed deviations for acceptance. Where such deviations require any changes to the structure, building works or services, these shall be provided by the Contractor without variation to the Contract. Apart from deviations listed, it will be taken that the materials and equipment proposed meets with all other specified requirements. Review of equipment and materials is undertaken on this basis, and that non-complying equipment/materials may be rejected at any time – even if already installed.
- 7 Submit amended As-Built documents as appropriate, highlighting changes since the previous submission.

2.5.1 SAMPLES

2.5.1.1 TIMING

Co-ordinate submissions of samples as listed in the Schedules. Do not cause delays by making late submissions or submitting inadequate samples.

2.5.1.2 QUANTITY

Submit a sample of each designated item and 2 copies of supporting documentation. Include ancillary items such as fasteners and mounting brackets. Amend and resubmit samples which are not accepted.

2.5.1.3 IDENTIFICATION

- 1 Identify the project, Contractor, their agents or supplier, manufacturer, applicable product, model number and options, as appropriate and include pertinent Specification references.
- 2 Include service connection requirements and product certification.
- 3 Identify non-compliances with project requirements, and characteristics that may be detrimental to successful performance of the completed work.

2.5.1.4 SAMPLE ACCEPTANCE AND RETENTION

- 1 Do not commence work affected by samples until the samples have been accepted. Submit further samples as necessary.
- 2 Keep approved samples in good condition on site, until Practical Completion.
- 3 Incorporate in the works samples, which have been approved for incorporation. Do not incorporate other samples.
- 4 Installed items will match accepted samples throughout the works.

2.6 MATERIALS, EQUIPMENT AND COMPONENTS

2.6.1 GENERAL

- 1 Provide new materials, equipment and components shall be new, of good quality and fit for purpose, selected for a reasonable service life. Do not provide without approval products that are obsolete, discontinued or about to be discontinued.
- 2 Identification of a proprietary item does not necessarily imply exclusive preference for the item so identified, but indicates a deemed to comply item.
- 3 Manufacturers: Provide equipment and associated accessories which are the products of established manufacturers regularly engaged in the manufacture of such equipment, who issue comprehensive rating data and certified test data on their products.
- 4 Consistency: For the whole quantity of each material or product use the same manufacturer or source and provide consistent type, size, quality and appearance.
- 5 Safety: Provide all necessary safety devices for the protection of personnel against injury and the protection of plant and equipment against damage including relief valves, belt guards, safety railing, effective earthing of electrical components, electrical interlocks, warning lights, and signs, alarms and local lighting. Provide permanent lifting eyes for equipment exceeding 40kg.
- 6 General: Select, if no selection is given, and transport, deliver, store, handle, protect, finish, adjust, prepare for use, and use manufactured items in accordance with the current written recommendations and instructions of the manufacturer or supplier.
- 7 Instructions: Submit the recommendations and instructions, and advise of conflicts with other requirements.
- 8 Project modifications: Advise of activities that supplement, or are contrary to, manufacturers or suppliers' written recommendations and instructions.

General Requirements

- 9 Product certification: If products must comply with product certification schemes, use them in accordance with the certification requirements.
- 10 Provide permanently fixed plates indicating manufacturer, model, serial number, capacity and electrical data for all equipment. Lettering height: 5mm maximum, 1.5mm minimum.
- 11 If materials or products are supplied by the manufacturer in closed or sealed containers

2.6.2 FINISHES AND VISUAL ELEMENTS

- 1 Surfaces shall be flat and free of noticeable distortion, warping and twisting.
- 2 Edges and corners accessible to users dressed to avoid cutting if touched.
- 3 Pressings shall be accurate in dimension and profile to ensure consistent mating and clearance gaps and pressing shape.
- 4 Mating surfaces between adjacent pressings - flush and accurate to produce smooth distortion free joint.
- 5 Provide protection against damage until Practical Completion.
- 6 Surface shall be clean, undamaged and free of stains.

2.6.3 CLEANING AND PROTECTION OF FINISHED WORK

During handling and installation of work at the site, clean and protect work in progress and adjoining work. Apply suitable protective covering on newly installed work where required to ensure freedom from damage or deterioration at a time of Completion; otherwise, clean and perform maintenance on newly installed work as frequently as necessary throughout remainder of construction period. Adjust and lubricate operable components to ensure operation without damaging effects.

2.6.4 PROHIBITED MATERIALS

- 1 Do not use any of the following materials:
 - a Lead where the metal or its corrosion products may be directly ingested, inhaled or absorbed, or any lead-based paints or primers.
 - b Chromate paint pigment or chromate water treatment.
 - c Urea formaldehyde foam or materials which may release formaldehyde in quantities which may be hazardous or irritant.
 - d Synthetic mineral fibres except in accordance with WorkSafe Australia National Standard and National Code of Practice.
 - e Materials in which chlorofluorocarbons, hydrochlorofluorocarbons or hexa-fluoroacetones have been used as a blowing agent.
 - f Chlorofluorocarbons.
 - g Polychlorinated biphenyls.
 - h Other substances generally known to be deleterious to health or safety or which would adversely affect the works.
 - i Any that would be contrary to achievement of a required project Green Star or other sustainability rating.

2.7 INSTALLATION

2.7.1 GENERAL

- 1 General: Carry out the work in a proper and professional manner.
- 2 Arrangement: Install equipment and services parallel or perpendicular to building elements. Organise reticulated services neatly. Provide for movement in both structure and service.
- 3 Movement and expansion: Provide expansion facilities in ductwork, piping, cables, cable trays and supports to accommodate thermal expansion and movement at structural expansion joints.
- 4 Protection: Protect equipment from weather and the ingress of dirt, moisture, vandalism and tampering.
- 5 Access: Provide access to all components requiring entry, inspection or maintenance.
- 6 If interruptions to supply of any service are required co-ordinate the shutdowns to the satisfaction of the Client representative, advise the occupants, minimise inconvenience, and advise when supply is reinstated. Arrange shutdowns out of hours.

2.7.2 SYSTEM INTEGRATION

- 1 Interconnect system elements so that the installations perform their designated functions.

2.7.3 SETTING OUT OPENINGS AND MAKING GOOD

- 1 The Drawings are diagrammatic only and unless otherwise stated shall not be used for determining the precise positions of equipment outlets and like features. The exact location of these shall be determined on site and from as-built drawings, shop drawings, site surveys and ground penetrating radar scans.
- 2 All chases, ducts, recesses and penetrations in structural elements not shown on the building and structural drawings shall be subject to approval.
- 3 Provide the precise location of all openings, fixings and similar items of work required for these works.
- 4 Determine requirements at such times as not to cause delay to the work.
- 5 Check all items to ensure correct positioning.
- 6 Wrongly located or omitted openings, fixings and similar items of work resulting from incorrect or lack of inter trade coordination shall be remedied and paid for under these works.

2.7.4 METALWORK

- 1 General: Use metalwork capable of transmitting the loads imposed, and sufficient to ensure the rigidity of the assembly without causing deflection or distortion of finished surfaces. Construct to prevent rattle and nuisance.
- 2 Metal separation: Prevent contact between electrolytically dissimilar metals, by using concealed insertion layers.
- 3 Edges and surfaces: Keep clean, neat and free from burrs and indentations. Remove sharp edges.

2.7.5 FIXING

- 1 Fix all plant to structure in approved manner, either directly or via secondary structure as required. Submit details of types of fixings, locations and loads for approval.
- 2 Fix only light weight items to non-structural building elements.
- 3 Do not pierce waterproof roofs, floors or walls with fixings.
- 4 Fasteners to comply with appropriate Australian / New Zealand Standards.

- 5 Use proprietary corrosion resistant fasteners capable of transmitting the loads imposed, and sufficient to ensure the rigidity of the assembly. Use metal expansion bolts for concrete and masonry. Do not use explosive charge fixings without approval.
- 6 Corrosion protection: Use corrosion resistant, chemically and electrolytically compatible fasteners. Provide insulating spacers where necessary.

2.8 CO-ORDINATION

Comprehensive co-ordination of all services and associated building work form part of the Works.

These responsibilities shall include:

- 1 Co-ordination of the works as required.
- 2 Co-ordination of advance procurement of equipment to meet programme.
- 3 Co-ordination of in-ground and in-ceiling services routes, including trenching, trays and pipework.
- 4 Management of the preparation and approval of fully co-ordinated drawings including penetration drawings.
- 5 Management of services testing and commissioning
- 6 Ensuring provision of all test results and required certifications to obtain Certificate of Occupancy from the relevant Building Surveyor.

2.8.1 CO-ORDINATION OF INSTALLATION WITH CEILING TILES AND FINISHES

Co-ordination shall be carried out to ensure that all building and services elements such as air ducts, outlets and light fittings, speakers and other ceiling elements are installed in a co-ordinated manner, such that all services can be connected and maintained as indicated on the Architectural Reflected Ceiling Plans.

2.9 SAFETY IN DESIGN

Safety in Design risk management shall form an integral part of these works and shall conform to the requirements of the Workplace Health and Safety (WHS) Act 2011 & WHS Regulations 2011.

The Safety in Design process shall conform to the State-relevant Compliance and Guidance documentation.

General principles of risk management to be followed shall be as AS/NZS ISO 31000 (2009) Risk Management – Principles and Guidelines.

The National Standard for Construction Work (NOHSC:1016) shall be used to align design risk management practice with construction risk management practice.

The Safety in Design Risk Assessment shall include but not be limited to:

- 1 Understand the range of work activities associated with the intended use of the building / structure as a workplace.
- 2 Identify hazards, assess risks (quantify and rank), identify control measures, implement control measures, review and prepare risk register prior to construction of work. Monitor and review throughout construction and operation. The process includes review and acknowledgement of residual risks identified and communicated by the designer as defined by the WHS Act and Regulations.
- 3 Elimination or where this is not reasonably practicable, minimisation of any risk to health and safety of any person. The process includes recommendation of design alternatives that will eliminate / reduce risks, recommendation of control measures for residual risks; organising participate and facilitating Safety in Design workshops as required.

General Requirements

- 4 Report in writing, on health and safety aspects and risks of the design identified. Implementation of control measures which shall be monitored and reviewed throughout construction and operation.
- 5 Provide at the conclusion of the project all residual risk information, in the form of a Safety in Design File and the related instruction and training as required to ensure an understanding of the safety aspects of the installation.
- 6 Where local regulatory practice differs from the above the more onerous shall be used as the basis for compliance.

3 PROJECT REQUIREMENTS AND DESCRIPTION OF WORKS

3.1 ROLES & RESPONSIBILITIES

3.1.1 OBLIGATIONS OF THE CONTRACTOR

- 1 In writing this Specification the Consulting Engineer expects and relies upon the Contractor to provide its own personnel and its trades possessing specialist trade expertise necessary to complete the works in accordance with the Contract documentation.
- 2 In addition, the Contractor has the following obligations:
 - a Complete the design while demonstrating compliance with the design intent.
 - b Raise in good time, issues requiring design input or clarification from the Consultant Engineer, particularly in respect to:
 - i interpretation of the Specification or drawings;
 - ii problems in complying with the Specification together with suggested alternative/substitutions;
 - iii matters in the Contractor's opinion, are omissions not discovered during the tender process, together with suggested alternatives/substitutions
 - c Certify compliance with Authority requirements.
 - d Pay all fees applicable to the works.
 - e Obtain all Authority permits and certificates to allow the progress of the work.
 - f Provide manufacturers and construction drawings of sufficient detail to allow proper fabrication, co-ordination and installation and a scale not less than 1:50; incorporating all post-tender changes to architectural and structural drawings.
 - g Provide samples and prototypes where specified.
 - h Provide manufacturer's factory test records (type tests and witnessed tests) and site installation inspection and sign-off certifying that the completed installation complies with the manufacturer's installation guidelines and requirements.
 - i Implement procedures to ensure that only competent tradesmen are used for the works.
 - j Be cognisant of the role of all relevant parties during the construction phase of the project and to assist them in the conduct of their duties wherever possible.
 - k Provide the completed Services installation that is fully and correctly commissioned and fine-tuned and, that is functional and operable efficiently in compliance with all requirements of this Specification.
 - l Certify compliance with Contract documents, including all variation instructions, at Practical Completion

3.1.2 CONTRACTOR'S DESIGN RESPONSIBILITIES

The Contract documents depict a design intent of the services and are not fully coordinated with the Architectural drawings or other services. These documents are provided for a performance-based design, documentation and installation contract. The Installer shall be responsible to fully develop the design in accordance with the requirements of the brief and the contract documents, carry out computations, attend meetings with the

Architect/Engineer, ECI Contractor and other trade Contractors. Completion of design drawings shall be undertaken prior to the development of shop drawings and construction drawings. The Contractor shall be responsible for the detailed design activities listed below, in addition to those activities under the Contract.

- 1 Detailed design and construction of fire hydrant and hose reel system.
- 2 Detailed design and construction of fire detection system and occupant warning system.
- 3 Detailed design of seismic restraints by qualified designers where required.
- 4 The Contractor shall be responsible for ensuring that the detailed design undertaken is fully co-ordinated and compatible with the remainder of the project design.
- 5 Check space requirements of equipment and services which are indicated diagrammatically in the Contract documents. Select equipment with dimensions to suit the available space.
- 6 Lay out equipment and services to be accessible for operation, maintenance and replacement and so as not to interfere with access to other installations. Make offsets as necessary.
- 7 Set out access ways 2.1m high and 1.0m wide (minimum) to all major plant clear of all obstructions, unless otherwise approved.
- 8 Co-ordinate the layout of plant and services with the building layout and structure, and with other plant and services.
- 9 Neatly group services, with separate layers for crossing services.
- 10 Interface details and coordination with other trades.
- 11 Size and location of penetrations in walls and floors and roofing.
- 12 Physical co-ordination of installation with other trades.
- 13 Provision of electrical loads to electrical trade and compare with design loads cable sizes and circuit protection devices, seeking design direction where discrepancies occur.
- 14 Details of electrical wiring and control diagrams of all equipment supplied showing all interconnections between equipment to enable the necessary wiring to be undertaken.
- 15 Thermal expansion accommodation and anchorage, including provision of bellows or bends, taking into account final installation details and consistent with specified requirements.
- 16 Mounting, support and fixing details, and fasteners, including any secondary structure which may be required.
- 17 Cable installation details and derating factors.
- 18 Settings for protection equipment, time delays, time switches etc.
- 19 Capacity, location, design and sizing of cable support, trunking and conduit systems.
- 20 Acoustic design or modification of actual selected equipment that may require change, to meet with the noise levels specified. Such change to approval of the Consultant Engineer. Specified levels to be achieved with all plant operating.
- 21 Hydraulic calculations, calculations of electrical demands, sound pressure levels and speech intelligibility as required.
- 22 Liaison with respective utility companies to confirm the location, capacity (size, pressures, flow, etc.) and method of connection of all tapings from the water supplies between the property and the town mains
- 23 Functional design, spatial allowances, equipment supports and servicing access requirements. Location of all plant access requirements shall be identified on the shop drawings
- 24 Final location of system components. Final locations to take into functional and architectural requirements. Location of all fire protection services plant and field devices shall be identified on the shop drawings.

- 25 Co-ordination of the works between other Works Packages including any alterations within the Contract. Alterations undertaken within the Contract are to not adversely affect other Works Packages.
- 26 Coordination of the construction of the installation.

3.1.3 COORDINATION

- 1 Comprehensive co-ordination of all services and associated building works forms part of the Works. Ensure that the detailed design undertaken is fully co-ordinated and compatible with the remainder of the project design. These responsibilities shall include:
 - a Checking of space requirements of equipment and services which are indicated diagrammatically in the Contract documents. Select equipment with dimensions to suit the available space. Prior to ordering equipment, plant, switchboards, etc. ensure that sufficient space is available, by reference to the Architectural drawings or by site measurement, to accommodate the actual size of the equipment being offered.
 - b Lay out equipment and services to be accessible for operation, maintenance, and replacement and so as not to interfere with access to other installations. Make offsets as necessary. Access ways shall be 2.1m high and 1.0m wide (minimum) to all major plant clear of all obstructions, unless otherwise approved.
 - c Co-ordination of the layout of plant and services with the building layout and structure, and with other plant and services. Co-ordination of the works includes but is not limited to mechanical, hydraulics, structural, acoustic, fire protection and architectural works prior to installation. The contractor shall satisfy themselves that any alterations within the Contract do not adversely affect other work. Neatly group services, with separate layers for crossing of services.
 - d Comprehensive checking of plant, systems, piping, cabling, etc. capacities and performance against the final arrangement of equipment, e.g. checking capacity of utility connections, checking current carrying capacities / voltage drop, checking for system interfaces and interference in electrical and BMS systems, etc.
 - e Arrangement and co-ordination of all interconnection testing with other services systems, e.g. fire mode tests, power failure testing, etc.
 - f Co-ordination of the construction of the installation.
 - g Physical co-ordination of installation with other trades.
 - h Interface details with other trades.
 - i Size and location of penetrations in walls and floors including those fabrics that do not form part of the works under contract, but which are necessary to realise the works under contract.
 - j Co-ordination of in-ground and in-ceiling services routes, including trenching, trays and pipework.
 - k Management of the preparation and approval of fully co-ordinated drawings including penetration drawings.
 - l Arrangement and management of inter services / integrated services testing and commissioning.
 - m provision of all test results and required certifications to obtain Certificate of Occupancy from the Building Surveyor.
 - n Co-ordination to ensure that all building and services elements such as air ducts, outlets and light fittings, speakers and other ceiling elements are installed in a co-ordinated manner, such that all services can be connected and maintained as indicated on the Architectural Reflected Ceiling Plans.
- 2 During the construction period the works require that a competent supervisor is provided on site responsible for the comprehensive co-ordination of all engineering services, including civil, structural, mechanical, hydraulics, fire and lift services. Responsibility for the proper direction, supervision, control and co-ordination of the work forms part of the works and shall assist all other trades in every way possible in meeting this responsibility.
- 3 Should work proceed without co-ordination being undertaken and authorisation by those with such responsibility and should other trades subsequently not be able to effectively proceed with their respective installation, the work initially undertaken shall be removed and reinstalled at the expense of the Contractor.
- 4 Any claims for additional costs or delays due to the lack of effective co-ordination by the various trades will be rejected.

3.1.4 *ALTERNATIVE PRODUCTS*

- 1 Should the part of the work be proposed to be done quicker, better or more effectively by substitution of materials or methods other than those specified, the details of such substitutions or alternatives shall be included in the Tender, including a comprehensive report including any performance and life cycle analysis in support of the proposals.
- 2 It is the contractor's responsibility to ensure that all equipment orders are made in a timely fashion to suit the project construction programme. The proposing of alternatives as a result of the contractor's failure to do so will not be accepted as a means for alternative submissions.
- 3 Where the words 'equal or approved' are used in this Technical Specification, permission may be requested to use a substitute for what is specified, provided it and the relevant manufacturer of the item certify in writing that the substitute is of equal, or better quality and effectiveness to that specified.
- 4 Submit for approval full details to allow checking that the alternative products comply with the Contract Documents. Provide a certificate for each alternate product verifying that the proposed alternative product complies with the contract documents. Include all compliance and non-compliance items related to Project Brief, Specifications and Contract Drawings. State if use of proposed alternatives will necessitate alteration to other parts of the works and include consequential costs.
- 5 Allow for additional work resulting from the utilisation of an approved alternative product, including additional or revised statutory approvals, changes to adjacent work, re-submission of shop drawings and any costs incurred by the Consultant Engineers in reviewing such alternative products. Allow for Consultant Engineer Review cost based on agreed Market hourly rates. These sums are to be paid irrespective of the alternatives being selected or not.
- 6 Proposals for substitution must include details of:
 - a Changes to adjacent work, if any.
 - b Cost differences.
 - c Life cycle cost differences
 - d Quality differences, Specification compliance and advantage to the Client.
 - e Lead time and program changes.
 - f Certificate of compliance.

3.2 STAGING OR SEQUENCING OF WORKS

- 1 Refer to the Principal's requirements for staging & sequencing of the works.
-

3.3 DESCRIPTION OF WORKS

Whether or not particular works are described in the Technical Specification all items and materials needed for the complete works are required and shall be installed unless clearly excluded.

The project includes the detailed design, supply, delivery, installation, commissioning, tuning, testing, placing into service, maintenance, warranty and defects liability of materials, labour and plant of the services systems in accordance with this Technical Specification. All components and systems shall be complete in every respect and tested and commissioned unless otherwise noted specifically.

3.3.1 *GENERAL*

- 1 Allow to decommission, make safe and remove any existing redundant services to be deleted within the area of works.
- 2 Site visit and familiarization with the site conditions and scope of work required under this Contract and making required allowances to complete the works.

- 3 All necessary negotiations including formal submissions as required with all Authorities/utilities having jurisdiction including obtaining of design approval and final certification of the installed systems by an independent certifier approved by the Authority. Payment of all fees associated with the authorities/utilities connections and permits for the installation
- 4 Provision of detailed safety in design analysis and ensure the installation is in compliance with the Site's safety in design proposal including all regulatory requirements.
- 5 Engagement and coordination with the Green Star / Sustainability Consultant including timely preparation of documents as required by them.
- 6 Collaboration and coordination with other trades.
- 7 Coordinate with the Electrical trade and advise final equipment selections and their power requirements prior to their ordering and installation of submains to respective switchboard to ensure correct submain cable selections. Any submains/sub circuit to secondary switchboards/control panels shall be included in the respective services (Mechanical, Hydraulics, Fire, Civil, Vertical Transportation) Work Packages
- 8 Preparation of "For Construction" and "Shop Drawings" coordinated with other services prior to commencement of works including services layouts at the same scale as the design drawings.
- 9 Provision of all samples and technical submissions.
- 10 Identification and coordination of penetrations, provision of fire/smoke sleeves and sealing of all wall, floor, ceiling penetrations to the required fire/smoke/acoustic rating.
- 11 Provision of vibration isolation for all rotating equipment.
- 12 Provision of seismic restraints, including design.
- 13 Provision of all painting and finishes including identification and labelling schedules,
- 14 Provision of As-Built drawings and O&M Manuals in compliance with a uniform overall Site Services Documentation System.
- 15 Provision of all training for staff in the operation and preventative maintenance of the systems. Training of the nominated building facility management staff to the level of understanding of day-by-day operation of the services systems
- 16 Statutory testing, servicing, maintenance, defect liability and warranty of all services systems/equipment installed under this Contract from the date of Practical Completion
- 17 Fine tuning commissioning of all systems during the defects liability period as per Green Star/NABERS requirements. The tuning shall involve inspection, check and all adjustments of all systems and controls every three (3) months after the date of Practical Completion. Detailed final recommissioning of all systems shall be carried out at the end of the defect liability period.

3.3.2 *INTER SERVICES WORKS*

- 1 Fire extinguishers for protection of switch rooms, electrical risers, electrical cupboards, comms rooms or cupboards.
- 2 Smoke detection system in communication rooms/cupboards, switch rooms, electrical risers, electrical cupboards.
- 3 Aspirated smoke detection system at top of lift shafts and in new trades building.
- 4 Interface with mechanical services switchboards for fire shutdown signal.
- 5 Interface with door controllers for open/close signals to operate for compartment integrity.
- 6 Interface with vertical transport controllers for fire mode signal.
- 7 Interface with BMS including HLI and LLI signals.

- 8 Cable tray risers installed within comms cupboards.

3.3.3 SYSTEMS

1 Building 19

- a Provide Fire detection system throughout level 3 as documented. Fire Detection Control and Indicating Equipment (FDCIE) is to be upgraded to facilitate connection of new devices. Existing panels networking to be maintained.
- b Provide Emergency Warning and Intercom System throughout level 3 as documented. New devices to be connected to existing and upgraded Emergency Intercom Control and Indicating Equipment. Cone speakers generally throughout to satisfy the requirements of AS1670.4. Horn speakers shall be provided to plant type areas. Warden Intercom Point (WIP) phones are to be provided as documented.
- c Provide Fire extinguishers, brackets and signage as required.

2 Building 21A Engineering

- a Provision of external hydrant extended from the existing campus fire hydrant water supply.
- b Provision of occupant warning speaker system upgrade, connected to the existing occupant warning system.
- c Provision of Fire extinguishers, brackets and signage as required.

3 Building 32

- a Allow to relocate a detector and speaker in each room to facilitate the installation of new full height partition within the room. Where relocation is not required, the cost associated with the works shall be refunded to the client at the agreed contract price.

4 Building 33

- a Provision of Hydrant and Hose reel system upgrade throughout the building as documented, extended from the existing internal fire hydrant water supply pipework.
- b Provide Fire detection system throughout the building as documented. Fire Detection Control and Indicating Equipment (FDCIE) is to be replaced with new Notifier AFP3030 to facilitate connection of new devices. Existing panels networking to be maintained.
- c Provide Occupant Warning System throughout the building as documented. New devices to be connected to new Emergency Warning Control and Indicating Equipment. Cone speakers generally throughout to satisfy the requirements of AS1670.1. Horn speakers shall be provided to plant type areas, and external areas.
- d Provide aspirating smoke detection (ASD) at the top of lift shafts. ASD to be connected to FDCIE and provide at least two (2) warning/alarm output levels of smoke contamination to the fire detection system.
- e Provide Fire extinguishers, brackets and signage as required to Level 1 and 2.

5 Building 34

- a Provide Fire extinguishers, brackets and signage as required.
- b Allow to remove and reinstate existing detectors and speakers to facilitate builders works to ceilings.

6 Building 36

- a Provision of external hydrant and new internal fire hose reels extended from the existing campus fire hydrant water supply.
- b Provision of relocation of existing FDCIE and network connection. Upgrade FDCIE to facilitate connection of new manual call points and electric alarm bells throughout the building. Existing panels networking to be maintained.

- c Provide Fire extinguishers, blankets, brackets and signage as required.

7 Building 76 Horticulture

- a Provision of external hydrants and new internal fire hose reels extended from the existing campus fire hydrant water supply. Provide weatherproof metal cabinets for fire hose reels installed in external areas.
- b Provide Fire extinguishers installed in weatherproof metal cabinets for fire extinguishers installed in external areas.

8 Building 92

- a Provide external hydrants and new internal fire hose reels extended from the existing campus fire hydrant water supply.
- b Provide Fire detection system throughout the building as documented. Fire Detection Control and Indicating Equipment (FDCIE) is to be a new Notifier AFP3030. Connect new FDCIE with existing campus detection network main fire panel in building 06.
- c Provide Occupant Warning System throughout the building as documented. New devices to be connected to new Emergency Warning Control and Indicating Equipment. Cone speakers generally throughout to satisfy the requirements of AS1670.1. Horn speakers shall be provided to plant type areas, carpark, and external areas.
- d Provide aspirating smoke detection (ASD) at the top of lift shafts and as documented in workshop areas. ASD to be connected to FDCIE and provide at least two (2) warning/alarm output levels of smoke contamination to the fire detection system.
- e Provide Fire extinguishers installed in tamperproof cabinets where accessible to people outside the building or in a location where they may be stolen. Provide weatherproof metal cabinets for fire extinguishers installed in external areas.

- 9 All fire services to be designed and installed in accordance with the requirements of the Fire Engineering Report.

- 10 Provision of certification, as installed drawings, block plans, logbooks, baseline data, and operation & maintenance manuals.

3.4 SHOP DRAWING AND CONSTRUCTION DRAWING REQUIREMENTS

- 1 The term shop drawings include manufacturing and For Construction drawings and shall mean complete drawings showing all details of fabrication assembly, installation and fixing of the items concerned, and shall include any necessary explanatory notes or specifications.
- 2 Prepare, and submit for review before commencing manufacture or installation, Construction Shop Drawings from which the subject works shall be built. Shop drawings shall be dimensioned, showing details of the fabrication, layout and installation of all plant and equipment, including relationship to building structure and other services. No factory or site work, other than manufacture of standard equipment or controls, shall proceed until such drawings have been reviewed and marked as satisfactory. Show all dimensions and note which are based on site measurement. Confirm dimensions on site prior to preparation of drawings. Allow for adjustments and fitting on site. Indicate whether site dimensions have been verified. Confirm site dimensions prior to fabrication. No extra charge or compensation will be considered for differences between actual dimensions and dimensions indicated on Contract Documents. Do not delay job progress.
- 3 Where plant and equipment are to be located within or above the structure, submit sketch plans at an early stage of the project to demonstrate that the spatial allocations are sufficient for effective installation, testing, commissioning and lifetime maintenance or replacement.

- 4 Co-ordinated penetration drawings shall be submitted in sufficient time for review. Any failure to submit requirements within sufficient time for revision of structural drawings, resulting in re-working of the structure or re-ordering of structural components shall be responsibility of Contractor and shall not constitute any claim for variation or delay. All penetration measures shall relate to grid lines, co-ordinates or relative levels.
- 5 Prepare and submit drawings of penetrations and “built-in” components in the same form and manner as described for the processing of shop drawings. All penetration measures shall relate to grid lines, co-ordinates or relative levels.

Provide drawings generally in accordance with the design drawings supplied with this specification.

Sheet sizes: Standard metric series, all sheets the same size. Plant room layout plans 1:50 scale, with sections and details at 1:20 scale. All services Floor plans at 1:100 scale, with detailed sections.

Drawings:

- 1 Access to and removal of all plant including locations and sizes of access doors to be installed in finished building surfaces.
- 2 Schematic and layout drawings of piping, tray and conduits, electrical and controls.
- 3 Switchboard and control panel layouts and schematics with terminal numbering.
- 4 Locations of all control sensors, valves and actuators.
- 5 Detail of all interfaces with other services and utilities.
- 6 Services co-ordination drawings for spatial co-ordination with building structure and other services with details for restricted locations.
- 7 Control logic diagrams with settings.
- 8 Foundations, plinths, chases, ducts, pits and penetrations through structure.
- 9 Layout and details of services cast in concrete. (for co-ordination and review by structural engineer /architect)
- 10 Structural support details and methods of fixing to structure.
- 11 Construction loadings (for structural engineer review).
- 12 Lifting point loads. (for structural engineer review)
- 13 Maintain current sets of drawings on site and progressively record variations which lead to creation of as-built drawings.

Please note when shop drawings review comments states “satisfactory subject to comments”, resubmit the revised drawings with comments incorporated within a period of no longer than 2 weeks. However, this should not stop the installation works.

3.5 BIM/REVIT REQUIREMENTS

- 1 Refer to the CQU design standards manual Rev29 All project documentation must follow the defined structure, and details are contained within the CQUniversity Drawing Standards (Appendix E).

3.6 SITE PROVISIONS & BUILDING WORKS IN CONNECTION

- 1 Include site provisions as well as Building Works in Connection BWIC with this service. The Contractor is responsible for demarcation of tasks and responsibilities between various trades in relation to these items to suit his/her delivery methodology.
- 2 Site Provision include but not limited to;
 - a Hoisting and Lowering:
 - b Scaffolding:
 - c Temporary Services – Water, Electricity:
 - d Site facilities including Sheds.
 - e Power & Fuel for Testing and Commissioning

3.6.1 *BUILDING WORKS IN CONNECTION*

Allow for the following;

- 1 Demolition:
 - a Services decommissioning, disconnection from supply, capping and sealing.
 - b Services demolition and removal.
- 2 Ground Works: Services trenches; excavation and backfilling.
- 3 Concrete:
 - a Chasing, coring, cutting and making good.
 - b Placing, casting in and protection of pipe sleeves and conduits.
 - c Bases and plinths for equipment (excluding galvanised steel edge surround to be provided as part of the Mechanical Services Works Package).
 - d Piers and beams under cooling towers, fluid coolers and adiabatic coolers.
 - e Water-proof curbs round floor penetration in plant rooms.
 - f Masonry:
 - g Chasing, coring, cutting and making good.
 - h Plant enclosures, masonry air ducts and service risers.
- 4 Structural Steel:
 - a Lifting beams.
 - b Platforms and walkways for maintenance to equipment.
- 5 Roofing:
 - a Openings and under flashings including over flashings
 - b Services works.
 - c Roof access walkways.
- 6 External Walls:

- a Metal louvre grilles complete with bird mesh for air intakes and exhausts.
- 7 Internal walls:
 - a Setout of cut-outs including cut-out and trimming of openings
- 8 Doors:
 - a Cut-outs in doors and trimming of openings
 - b Undercutting of doors for ventilation purposes.
 - c Access doors and hatches for maintenance of Services equipment.
- 9 Ceilings:
 - a Removal and replacement of ceiling tiles.
 - b Co-ordination of penetrations.
 - c Ceiling access panels to equipment, valves, dampers and duct access panels above solid ceilings.
 - d Sealed air Plenums for smoke exhaust systems.
- 10 Dry wall/ Plenum Construction:
 - a Sealed air plenums for supply and exhaust systems where show on the drawings, including fire rating and thermal insulation where required.
 - b Co-ordination of penetrations.
 - c Vapour barriers within walls where required.

3.7 SERVICES INTERFACE WORKS

Carry out the works outlined under all contract documents including this Specification to achieve a fully operational installation including all required interfaces between various systems. Whilst the responsibility for interface works and division of works is solely with the Contractor the following information is provided for guidance only. The Contractor has full control to alter and modify the trade allocation/demarcation of interface works to suit the Contractor's overall programme and delivery methodology in order to achieve a fully integrated, interfaced and operational whole services installations, systems and sub-systems.

3.7.1 TRADE PACKAGE INTERFACES

Table below indicates key areas where different work packages will interface with each other. Column 1 indicates the disciplines/work packages identified and columns 2 & 3 outline the key interface works between these two work packages.

	DISCIPLINE 1	DISCIPLINE 2
Fire 1 /Mech2	— General Fire Alarm cabling to (MDBs) to initiate shut down and reset of the respective Mechanical Plants. — Smoke detection in air plenums, ducts and AHU's if required. — High Level Interface (BACnet protocol) of all alarm and indication signals from the FDCIE to the termination point adjacent to the BAS server. Allow for 100 individual signals. Provision of	— Termination of fire alarm cabling at MSSBs to initiate smoke hazard management systems operation on GFA (inclusive of shut down operation and any reset requirements). — Internal wiring within the MSSB's / Damper Control Boards including all necessary control gear, terminal strips, etc. to provide a complete and operative system.

	DISCIPLINE 1	DISCIPLINE 2
	HLI cabling from the FDCIE to BAS/BMS panel. Assist with the development of the fire mode matrix.	- Access panels in ducts requiring smoke detectors to be fitted - Connection of HLI (BACnet Protocol) and cabling to BAS terminal point - Fire mode matrix. - All required software, hardware, functional descriptions, programming, trending, alarming, testing and commissioning, ITPs, warranty, maintenance, DLP associated with BMS / BAS interfaces and all associated graphic screens to depict such interfaces clearly.
Fire1/ Elec 2	- Provision of Volt free contact output for transmission of a General Fire Alarm to the Security Control system to allow for release of door locking on fire alarm. - Provision of cabling from FDCIE to respective security control panel to allow for release of door locking on fire alarm. - Terminate 240V supply cable to CIE power supplies. - Provision of GFA relay output signal to the background music system to mute BGM system. - Provision of smoke detection system in security control centre, electrical switch rooms, risers, switchboard cabinets, distributor rooms, and communications room. - Provision of fire extinguishers for protection of switch rooms, electrical switchboards, risers, security and computer rooms. - Provision of cable tray risers within comms cupboards.	- Termination of GFA cabling to security control system to allow for release of door locking on fire alarm. - Receive and terminate cabling for BGM mute signal on the BGM system and mute speakers during fire system tones or announcements - Provision of dedicated 240V power supply to fire detection system FDCIE. Final connection by Fire Protection Works Package. - Provision of dedicated 240V power supply to evacuation system EWCIE. Final connection by Fire Protection Works Package. - Provision of signage to all circuit breakers associated with power supply to fire protection equipment. - Provision of identification of all switchboards in accordance with the electrical services Scope of Works. - Provision of space in cable tray riser cupboards accommodate fire detection and evacuation systems cable containment.
Fire1 /VT2	- Provision of fire detection system in lift shafts above the controller. - Provision of GFA relay output signal to the VT controller for fire mode.	Receive and terminate cabling for GFA signal.

4 DESIGN AND PERFORMANCE CRITERIA

4.1 GENERAL

- 1 The following standards are applicable throughout the project:
- 2 National Construction Code (2022) (Referred throughout this specification as NCC) including all relevant Australian Standards called in the NCC
- 3 Relevant Occupation/Workplace Health and Safety legislation
- 4 The Environmental Protection Authority
- 5 WorkCover
- 6 Any other Authority having jurisdiction over all or part of the installation to ensure that the equipment and installation, when manufactured and installed, will comply with the rules and regulations.
- 7 Refer to section 1.4 for Additional and Relevant documentation list.

The following standards are applicable throughout the project as a minimum. Where items of equipment are required the relevant Australian Standard to the equipment will apply. If clarification required, the Contractor shall seek clarification at the time of tender.

Where Australian Standards and Codes do not exist, the appropriate British Standard or Code shall apply.

Details of documents applicable to this:

4.2 AUSTRALIAN/NEW ZEALAND STANDARDS & CODE (FIRE SERVICES)

STANDARD / CODE	TITLE
AS ISO 7.1	Dimensions, tolerances and designation
AS ISO 1000	The international system of units (SI) and its application
AS 1074	Steel tubes and tubulars for ordinary service
AS/NZS 1125	Conductors in insulated electric cables and flexible cords
AS/NZS 1163	Cold-formed structural steel hollow sections
AS/NZS 1167.1	Welding and brazing - Filler metals - Filler metal for brazing and braze welding
AS/NZS 1167.2	Welding and brazing - Filler metals – Filler metal for welding
AS 1170.4	Structural Design Actions - Earthquake actions in Australia
AS/NZS 1200	Pressure equipment
AS/NZS 1221	Fire hose reels
AS 1284.1	Electricity metering - General purpose induction watthour meters

STANDARD / CODE	TITLE
AS 1318	Use of colour for the marking of physical hazards and the identification of certain equipment in industry (known as the SAA Industrial Safety Colour Code)
AS 1319	Safety signs for the occupational environment
AS 1345	Identification of the contents of pipes, conduits and ducts
AS 1349	Bourdon tube pressure and vacuum gauges
AS 1359.0	Rotating electrical machines - General requirements - Introduction and list of Parts
AS/NZS 1359.5	Rotating electrical machines - General requirements - Three-phase cage induction motors – High efficiency and minimum energy performance standards requirements
AS 1359.30	Rotating electrical machines - General requirements - Preferred outputs and frame sizes
AS 1359.31	Rotating electrical machines - General requirements - Three-phase induction motors - Operation on unbalanced voltages
AS 1359-101	Rotating electrical machines - General requirements - Rating and performance
AS 1359.114	Rotating electrical machines - General requirements - Vibration measurements and limits
AS 1397	Continuous hot-dip metallic coated steel sheet and strip – coatings of zinc and zinc alloyed with aluminium and magnesium
AS 1428.1	Design for access and mobility – General requirements for access – New building work
AS 1428.2	Design for access and mobility – Enhanced and additional requirements – Buildings and facilities
AS 1428.3	Design for access and mobility – Requirements for children and adolescents with physical disabilities
AS 1432	Copper tubes for plumbing, gas fitting and drainage applications
AS/NZS 1477	PVC pipes and fittings for pressure applications
AS 1530.1	Methods for fire tests on building materials, components and structures – Combustibility test for materials
AS 1530.2	Methods for fire tests on building materials, components and structures – Test for flammability of materials
AS/NZS 1530.3	Methods for fire tests on building materials, components and structures - Simultaneous determination of ignitability, flame propagation, heat release and smoke release
AS 1530.7	Methods for fire tests on building materials, components and structures – Smoke control assemblies – Ambient and medium temperature leakage test procedure
AS/NZS 1554.1	Structural steel welding - Welding of steel structures
AS/NZS 1554.6	Structural steel welding - Welding stainless steels for structural purposes
AS 1603.1	Automatic fire detection and alarm systems – Heat detectors
AS 1603.2	Automatic fire detection and alarm systems – Point type smoke detectors
AS 1603.3	Automatic fire detection and alarm systems – Heat alarms
AS 1603.5	Automatic fire detection and alarm systems – Manual call points

STANDARD / CODE	TITLE
AS 1603.7	Automatic fire detection and alarm systems – Optical beam smoke detectors
AS 1603.8	Automatic fire detection and alarm systems – Multi-point aspirated smoke detectors
AS 1603.11	Automatic fire detection and alarm systems – Visual warning devices
AS 1603.13	Automatic fire detection and alarm systems – Duct sampling units
AS 1603.14	Automatic fire detection and alarm systems – Point type carbon monoxide (CO) fire detectors
AS 1603.15	Automatic fire detection and alarm systems – Remote indicators
AS 1603.16	Automatic fire detection and alarm systems – In situ testers
AS 1603.17	Automatic fire detection and alarm systems – Warning equipment for people with hearing impairment
AS 1604.1	Specification for preservative treatment – sawn and round timber
AS 1628	Water supply - Metallic gate, globe and non-return valves
AS 1657	Fixed platforms, walkways, stairways and ladders - Design, construction and installation
AS/NZS 1665	Welding of aluminium structures
AS/NZS 1668.1	The use of ventilation and air-conditioning in buildings - Fire and smoke control in multi-compartment buildings
AS 1670.1	Fire detection, warning, control and intercom systems – Safety design, installation and commissioning - Fire
AS 1670.3	Fire detection, warning, control and intercom systems – Safety design, installation and commissioning – Fire alarm monitoring
AS 1670.4	Fire detection, warning, control and intercom systems – Safety design, installation and commissioning – Emergency warning and intercom systems
AS 1670.6	Fire detection, warning, control and intercom systems – Safety design, installation and commissioning – Smoke alarms
AS 1674.1	Safety in welding and allied processes - Fire precautions
AS 1674.2	Safety in welding and allied processes - Electrical
AS/NZS 1716	Respiratory protective devices
AS 1722.2	Pipe threads of Whitworth form – Fastening pipe threads
AS 1796	Certification of welders and welding supervisors
AS/NZS 1841.1	Portable fire extinguishers – General requirements
AS/NZS 1841.2	Portable fire extinguishers – Specific requirements for water type extinguishers
AS/NZS 1841.3	Portable fire extinguishers – Specific requirements for wet chemical type extinguishers
AS/NZS 1841.4	Portable fire extinguishers – Specific requirements for foam type extinguishers
AS/NZS 1841.5	Portable fire extinguishers – Specific requirements for powder type extinguishers
AS/NZS 1841.6	Portable fire extinguishers – Specific requirements for carbon dioxide type extinguishers

STANDARD / CODE	TITLE
AS/NZS 1841.7	Portable fire extinguishers – Specific requirements for vaporizing type extinguishers
AS/NZS 1841.8	Portable fire extinguishers – Specific requirements for non-rechargeable type extinguishers
AS 1851	Routine service of fire protection systems and equipment
AS 1852.441	International electrotechnical vocabulary – Switchgear, control gear and fuses
AS 1905.2	Components for the protection of openings in fire resistant walls – Fire-resistant roller shutters
AS 1939 Supplement 1	Degrees of protection provided by enclosures for electrical equipment (IP Code) - Wallchart 1 (Supplement 1 to AS 1939-1990)
AS 1939 Supplement 2	Degrees of protection provided by enclosures for electrical equipment (IP Code) - Wallchart 2 (Supplement 2 to AS 1939-1990)
AS 1940	The storage and handling of flammable and combustible liquids
AS/NZS 2053.1	Conduits and fittings for electrical installations - General requirements
AS/NZS 2053.2	Conduits and fittings for electrical installations – Rigid plain conduits and fittings of insulating material
AS/NZS 2053.3	Conduits and fittings for electrical installations - Rigid plain conduits and fittings of fibre-reinforced concrete material
AS/NZS 2053.4	Conduits and fittings for electrical installations – Flexible plain conduits and fittings of insulating material (IEC 60614-2-5 Ed. 1.0 (Bilingual 1992))
AS/NZS 2053.5	Conduits and fittings for electrical installations – Corrugated conduits and fittings of insulating material
AS/NZS 2053.7	Conduits and fittings for electrical installations - Rigid metal conduits and fittings
AS/NZS 2053.8	Conduits and fittings for electrical installations – Flexible conduits and fittings of metal or composite material (IEC 60614-2-5 Ed. 1.0 (Bilingual 1992))
AS 2220	Emergency warning and intercommunication systems in buildings
AS 2239	Galvanic (sacrificial) anodes for cathodic protection
AS/NZS 2243.3	Safety in laboratories – Microbiological safety and containment
AS/NZS 2243.6	Safety in laboratories – Plant and equipment aspects
AS/NZS 2280	Ductile iron pressure pipes and fittings
AS/NZS 2310	Glossary of paint and painting terms
AS 2419.1	Fire hydrant installations – System design, installation and commissioning
AS 2419.2	Fire hydrant installations – Fire hydrant valves
AS 2419.3	Fire hydrant installations – Fire brigade booster connections
AS 2441	Installation of fire hose reels

STANDARD / CODE	TITLE
AS 2444	Portable fire extinguishers and fire blankets - Selection and location
AS 2467	Maintenance of electrical switchgear
AS 2625.1	Mechanical vibration - Evaluation of machine vibration by measurements on non-rotating parts - General guidelines
AS 2670.1	Evaluation of human exposure to whole-body vibration – General requirements
AS 2676.1	Guide to the installation, maintenance, testing and replacement of secondary batteries in buildings - Vented cells
AS 2676.2	Guide to the installation, maintenance, testing and replacement of secondary batteries in buildings - Sealed cells
AS 2700	Colour standards for general purposes
AS 2738	Copper and copper alloys - Compositions and designations of refinery products, wrought products, ingots and castings
AS 2784	Endless wedge belt and V-belt drives
AS 2832.1	Cathodic protection of metals - Pipes and cables
AS 2832.1	Cathodic protection of metals – Pipes and cables
AS 2832.4	Cathodic protection of metals – Internal surfaces
AS 2848.1	Aluminium and aluminium alloys - Compositions and designations - Wrought products
AS 2941	Fixed fire protection installations - Pump set systems
AS/NZS 3000	Electrical installations (known as the Australian/New Zealand Wiring Rules)
AS/NZS 3003	Electrical installations – Patient areas
AS/NZS 3008.1.1	Electrical installations - Selection of cables - Cables for alternating voltages up to and including 0.6/1 kV - Typical Australian installation conditions
AS 3011.1	Electrical installations - Secondary batteries installed in buildings - Vented cells
AS 3011.2	Electrical installations - Secondary batteries installed in buildings - Sealed cells
AS/NZS 3013	Electrical installations – Classification of the fire and mechanical performance of wiring system elements
AS/NZS 3100	Approval and test specification - General requirements for electrical equipment
AS 3111	Approval and test specification - Miniature overcurrent circuit-breakers
AS/NZS 3191	Electric flexible cords
AS/NZS 3439.1	Low-voltage switchgear and control gear assemblies - Type-tested and partially type-tested assemblies
AS/NZS 3439.2	Low-voltage switchgear and control gear assemblies - Particular requirements for busbar trunking systems (busways)

STANDARD / CODE	TITLE
AS/NZS 3439.3	Low-voltage switchgear and control gear assemblies - Particular requirements for low-voltage switchgear and control gear assemblies intended to be installed in places where unskilled persons have access for their use – Distribution boards
AS/NZS 3439.4	Low-voltage switchgear and control gear assemblies - Particular requirements for assemblies for construction sites (ACS)
AS/NZS 3500.0	Plumbing and drainage – Glossary of terms
AS/NZS 3500.1	Plumbing and drainage – Water services
AS/NZS 3500.1.1	Plumbing and drainage – Water supply – Performance requirements
AS/NZS 3500.2	Plumbing and drainage – Sanitary plumbing and drainage
AS/NZS 3500.2.1	Plumbing and drainage – Sanitary plumbing and drainage - Performance requirements
AS/NZS 3504	Fire blankets
AS/NZS 3518	Acrylonitrile butadiene styrene (ABS) compounds, pipes and fittings for pressure applications
AS 3715	Metal finishing - Thermoset powder coating for architectural applications of aluminium and aluminium alloys
AS/NZS 3750	Paints for steel structures
AS 3768	Guide to the effects of temperature on electrical equipment
AS 3772	Pre-engineered fire protection systems for cooking equipment
AS 3786	Smoke alarms using scattered light, transmitted light or ionisation
AS 3865	Calculation of the effects of short-circuit currents
AS 3996	Access covers and grates
AS 4029.1	Stationary batteries - Lead-acid - Vented type
AS/NZS 4029.2	Stationary batteries – Lead acid – Valve-regulated type
AS 4029.3	Stationary batteries - Lead-acid - Pure lead positive pasted plate type
AS 4041	Pressure piping
AS 4072.1	Components for the protection of openings in fire-resistant separating elements - Service penetrations and control joints
AS 4178	Electromagnetic door holders
AS/NZS 4251.1	Electromagnetic compatibility (EMC) - Generic emission standard - Residential, commercial and light industry
AS/NZS 4417.3	Regulatory compliance mark for electrical and electronic equipment - Specific requirements for electromagnetic compatibility regulatory applications

STANDARD / CODE	TITLE
AS 4428.0	Fire detection, warning, control and intercom systems – Control and indicating equipment – General requirements and test methods
AS 4428.1	Fire detection, warning, control and intercom systems – Control and indicating equipment – Fire
AS 4428.3	Fire detection, warning, control and intercom systems – Control and indicating equipment – Fire brigade panel
AS 4428.4	Fire detection, warning, control and intercom systems – Control and indicating equipment – Intercommunication systems for emergency purposes
AS 4428.5	Fire detection, warning, control and intercom systems – Control and indicating equipment – Power supply units
AS 4428.6	Fire detection, warning, control and intercom systems – Control and indicating equipment – Alarm signalling equipment
AS 4428.6	Fire detection, warning, control and intercom systems – Control and indicating equipment – Air-handling fire mode control panel
AS 4428.10	Fire detection, warning, control and intercom systems – Control and indicating equipment – Alarm investigation
AS 4428.16	Fire detection, warning, control and intercom systems – Control and indicating equipment – Emergency warning control and indicating equipment
AS 4506	Metal finishing - Thermoset powder coatings
AS 4594.0	Internal combustion engines - Performance - Application
AS 4594.1	Internal combustion engines - Performance - Standard reference conditions, declarations of power, fuel and lubricating oil consumption and test methods
AS 4594.3	Internal combustion engines - Performance – Engines for land, rail-traction and marine use – Test measurements
AS 4594.4	Internal combustion engines - Performance - Engines for land, rail-traction and marine use - Speed governing
AS 4594.5	Internal combustion engines - Performance - Engines for land, rail-traction and marine use - Torsional vibrations
AS 4594.6	Internal combustion engines - Performance - Engines for land, rail-traction and marine use - Overspeed protection
AS 4594.7	Internal combustion engines - Performance - Engines for land, rail-traction and marine use – Codes for engine power
AS/NZS 4680	Hot-dipped galvanized (zinc) coatings on fabricated ferrous articles
AS 4702	Polymeric cable protection covers
AS/NZS 4766	Polyethylene storage tanks for water and chemicals
AS/NZS 4791	Hot-dip galvanized (zinc) coatings on ferrous open sections, applied by an in-line process

STANDARD / CODE	TITLE
AS/NZS 4792	Hot dip galvanized (zinc) coatings on ferrous hollow sections, applied by a continuous or a specialized process
AS ISO 8601	Data elements and interchange formats - Information interchange - Representation of dates and times
CIBSE	Commissioning Codes
IEC 60085	Electrical insulation – Thermal evaluation and designation
NCC	Section J

4.3 SEISMIC RESTRAINTS

4.3.1 DESIGN STANDARD

- 1 NCC - Building Code of Australia.
- 2 AS1170.4 - Structural design actions - Earthquake actions in Australia

4.3.2 DESIGN PARAMETERS

The following design parameters shall be used to design the Engineering Services equipment and fastenings as per AS1170.4:

SEISMIC DESIGN PARAMETERS	
Importance Level (IL)	TBC
Site Sub-Soil Class	TBC
Hazard Factor (Z)	TBC
Probability Factor (KP)	TBC
Earthquake Design Category (EDC)	TBC

Design and install the scope of works to withstand the earthquake forces determined from the above design criteria in accordance with the requirements of the BCA and AS1170.4.

4.4 DESIGN CRITERIA FIRE SERVICES

Design criteria presented herein form the basis for the design of the works and for any alternative proposals submitted.

A1 General

ITEM	DESIGN CRITERIA
Fire Hydrant System	AS2419.1-2021, BCA as applicable and QFD operational requirements Full floor coverage by the use of 30 or 60m long hoses as applicable for internal and external hydrants. Simultaneous operation of three (3) hydrants each discharging 10l/sec @ 700 kPa when boosted by fire brigade.

ITEM	DESIGN CRITERIA
	Simultaneous operation of three (3) hydrants each discharging 5l/sec @ 700 kPa when pumped by onsite pumpset supplied by town's main.
Fire Hose Reels (FHR)	AS2441-2005, BCA as applicable Full floor coverage using 36m long hoses. Water supplied by fire hydrant system. Provide RPZD on fire hose reel water supply line where the FHR is located on Ground level of New Trade Building, and Horticulture Building, and within 36m of the Building 36 kitchen.
Fire Hydrant / Hose reel pipework and fittings Working Pressures	Galvanised black mild steel 1700 kPa Hot dipped galvanised black mild steel 1700 kPa Black mild steel 1700 kPa PE100 PN16 1700 kPa
Fire Detection and Alarm System All Enclosed Rooms	AS1670.1-2018 (A1), AS1603, AS/NZS 3000, BCA as applicable. Point detectors installed and spaced to AS1670.1
Fire Detection Control and Indicating Equipment (FDCIE)	AS7240.2-2018 Addressable Microprocessor Type
Emergency Warning Control and Indicating Equipment (EWCIE)	AS4428.16 (2015 or 2020)
Occupant Warning System	AS1670.1-2018, BCA as applicable Occupant warning system to comprise of Speakers utilising an 'evacuate' tone signal from tone generators and amplifier located in the EWCIE. Noise levels: — Maximum – 105 dBA — Minimum – 75dBA and 10dBA above the noisiest background sound pressure level Visual Alarm Devices: AS7240.23, BCA as applicable — Installed within high ambient noise areas with a background noise level exceeding 85 dBA — Install in areas designated for hearing impaired persons. — Install in areas where hearing protection devices are required to be worn.
Emergency Warning and Intercom System	AS1670.4-2018, BCA as applicable Warning system to comprise of Speakers utilising an 'alert' and 'evacuate' tone signal from tone generators and amplifier located in the EICIE. Warden Intercom Point phones and MCP's located on each level.

ITEM	DESIGN CRITERIA
	Noise levels: — Maximum – 105 dBA — Minimum – 75dBA and 10dBA above the noisiest background sound pressure level Visual Alarm Devices: AS7240.23, BCA as applicable — Installed within high ambient noise areas with a background noise level exceeding 85 dBA — Install in areas designated for hearing impaired persons. Install in areas where hearing protection devices are required to be worn.
Portable Fire Extinguishers	AS2444-2001, BCA as applicable
Fire Protection Pump Sets	NCC, AS 2941-2013
Existing Hydrant System	Existing Town main supply
Existing Characteristics	Existing Largest system duty 900L/min @ 700 kPa
Portable Extinguishers	NCC, AS2444
— Shopping Centre / Carpark	4.5 kg (6A:60B:E) dry chemical powder
— Electrical Rooms	5 kg (10B:E) carbon dioxide
— Plant Rooms	4.5 kg (6A:60B:E) dry chemical powder
— Major Switchboards and MSSB	5 kg (10B:E) carbon dioxide
— Kitchens	7 L (3A:4F) Wet Chemical
	4.5 kg (6A:60B:E) dry chemical powder & Fire Blanket
Maintenance/Verification	AS1670.1, AS1670.4, AS2419.1, AS2941, AS2441, AS2444, AS1851, NCC as applicable, Building Regulation 2006.

The above general design criteria are basically the deemed to satisfy criteria and shall be where required amended to suit the final Fire Safety Engineering Report.

4.4.1 ZONING DETAILS

4.4.1.1 FIRE ZONES

Install Fire Protection Systems to provide separate general fire alarm signal (fire detection and pressure/flow switches – as applicable) to the Fire Detection Control and Indicating Equipment from:

- 1 Each floor area of building.

4.4.1.2 ALARM CALL ZONING

Call to Fire Brigade via Alarm Signalling Equipment (ASE)

- 1 **FDCIE CALL** – General Fire Alarm transmission (Fire Detector/Break Glass Alarm)

4.5 ACOUSTIC

PARAMETER	DESIGN CRITERIA
Maximum Noise Levels	Refer to Schedules / project specific requirements

4.6 VIBRATION

PARAMETER	DESIGN CRITERIA
Maximum Vibration Levels	Tactile structure vibration levels due to plant not to exceed the lower end of the range specified in AS 2670.2.

5 PAINTING, PROTECTION AND IDENTIFICATION

5.1 GENERAL REQUIREMENTS

- 1 Select painting and finishes in accordance with the area environmental conditions. (i.e. Surface preparation, Degreasing)
 - 2 Review the risk assessment and statutory and authority requirements completed as part of the system testing, commissioning and handover requirements and provide labelling and signage as determined.
 - 3 Where exposed to view paint exposed equipment, cable trays, ductwork and pipework
 - 4 Where ductwork, pipework, cable trays exposed to view is not specified to be painted, all factory and installation markings shall be removed.
 - 5 All black steel surfaces and surfaces subject to corrosion shall be primed with at least two coats of primer and protected.
 - 6 Protect all items of work during dispatch and whilst on site during progress of installation.
 - 7 Fabricated pipe and steelwork shall be delivered to site prime coated
 - 8 Concealed pipe work to be prime coated
 - 9 If exposed to view, paint new services and equipment including in plant rooms, except chromium, anodised aluminium, GRP, UPVC, stainless steel, non-metallic flexible materials and normally lubricated machined surfaces. Repaint proprietary items only if damaged.
 - 10 Do not combine paints from different manufacturers in a paint system.
 - 11 Remove or protect fixtures, equipment, surfaces and labels before starting to paint, and re-instate after completion of painting.
 - 12 All internal surfaces of air outlets and openings exposed to view shall be painted matt black
 - 13 Repair factory finishes if damaged with identical finishes.
 - 14 Finish visible joints made by welding, brazing or soldering using methods appropriate to the class of work (including grinding or buffing) before further treatment such as filling, painting, galvanising or electroplating.
 - 15 If galvanised surfaces have been cut or welded after galvanising, prime the affected area using zinc rich organic binder AS/NZS 2312.
 - 16 Before applying coatings to metalwork, complete cutting, drilling and other fabrication, remove all grease, oil and other contaminants and prepare surfaces to AS 1627.0
 - 17 Use relevant Australian Standards for the following finishes; Galvanising, Electroplating, Anodising, Thermoset powder coating, Two-pack liquid coating, Air-drying enamel, Stoving enamel
-

5.2 SUBMISSIONS

- 1 Provide the following submissions or samples:
 - a Proposed Labelling Systems. (Engrave labels excluding pipe markers on red traffolyte with white lettering. Locate the labels as close as possible to the equipment it identifies, using lettering of a minimum 10mm high.)

- b Schedule of statutory, maintenance and operating signage.
- c Valve Identification

5.3 CORROSION PROTECTION

5.3.1 GENERAL

- 1 Apply corrosion protection measures to all metallic surfaces.
- 2 Select an appropriate protective coating system to suit atmospheric corrosivity category as defined in referenced Standards/Codes.
- 3 Fastenings are to be protected to have a corrosion resistance equivalent to the parts fastened.
- 4 All welding is to be completed before any corrosion protection treatment is applied.

ATMOSPHERIC CORROSIVITY CATEGORY	TYPICAL EXTERIOR ENVIRONMENTS	TYPICAL INTERIOR ENVIRONMENT
C1: Very Low	Few alpine regions	Heated or air-conditioned buildings, such as most of commercial buildings. Unheated or non-air-conditioned buildings in semi-sheltered locations remote from marine or industrial influence
C2: Low	Dry, rural areas as well as other regions remote from the coast or sources of pollution. Typical areas occur in arid and rural inland regions, most inland cities and suburbs of cities on sheltered bays.	Unheated or non-air-conditioned buildings, where some condensation may occur, such as warehouses and sports halls, can be in this category. Proximity to the coast is an important factor.
C3: Medium	Coastal areas with low salinity	Food processing plants, laundries, breweries, printing works, dairies
C4: High	Seashore (calm)	Swimming pools, dye works, paper manufacturers, foundries, smelters and chemical processing plants, livestock buildings
C5-I: Very High (Industrial)	Seashore /offshore	-
C5-M: Very High (Marine)	Shoreline (severe surf)	Adjacent to acidic processes
T: Inland Tropical	Non-coastal tropics	-

5.3.2 PROTECTION FOR STEEL SURFACES

- 1 Apply corrosion protection to meet the performance criteria scheduled in the table below.
- 2 Standards of surface preparation procedures and of corrosion treatments are to meet the specified criteria and the referenced Standards/Codes for the relevant application.

- 3 Surface preparation procedures and coating systems for corrosion protection are to meet all relevant requirements of the referenced Standards/Codes.

ATMOSPHERIC CORROSIVITY CATEGORY	SURFACE PREPARATION	COATING SYSTEM
C1: Very Low	Class Sa 2½ Blast Cleaning ‘Very thorough’ (commonly called ‘near white’) blast cleaning which removes millscale, rust and foreign particles to the extent that only traces remain.	1 st Coat – zinc phosphate alkyd primer, 75 micron thick 2 nd Coat – full gloss alkyd finish, 50 micron thick 3 rd Coat – full gloss alkyd finish, 50 micron thick Or Factory applied zinc primer and thermoset powder coated finish
C2: Low	Class Sa 2½ Blast Cleaning	1 st Coat – zinc phosphate alkyd primer, 75 micron thick 2 nd Coat – full gloss alkyd finish, 50 micron thick 3 rd Coat – full gloss alkyd finish, 50 micron thick Or Factory applied zinc primer and thermoset powder coated finish
C3: Medium	Class Sa 2½ Blast Cleaning	1 st Coat – Organic zinc primer, 75 micron thick 2 nd Coat – polyurethane, 100 micron thick Or Factory applied zinc primer and thermoset powder coated finish
C4: High	Class Sa 2½ Blast Cleaning	1 st Coat – Organic zinc primer, 75 micron thick 2 nd Coat – polyurethane, 100 micron thick Or Factory applied zinc primer and thermoset powder coated finish
C5-I Very High Industrial	Manufacturer’s recommendation	To approval
C5-M Very High Marine	Class Sa 2½ Blast Cleaning	1 st Coat – Organic zinc primer, 75 micron thick 2 nd Coat – polyurethane, 100 micron thick Or Factory applied zinc primer and thermoset powder coated finish
T Inland Tropical	Class Sa 2½ Blast Cleaning	One coat inorganic zinc coating, 75 micron thick Or zinc or aluminium Heat spray metallic coating.

5.3.3 PROTECTION FOR NON-FERROUS METALLIC SURFACES

- Standards of surface preparation procedures and of corrosion treatments are to meet the specified criteria and the referenced Standards/Codes for the relevant application.
- Surface preparation procedures and coating systems for corrosion protection are to meet all relevant requirements of the referenced Standards/Codes.

5.4 PAINTING

5.4.1 LOW VOC

- All painted surfaces to use low-VOC paints.
- VOCs are to be in accordance with The Australian Environmental Labelling Association, Inc. Standard No: AELA 23-2005 'Australian Voluntary Environmental Labelling Standard Architectural and Protective Coatings'.
Conformance with VOC levels listed in this standard (refer to table below) shall be demonstrated by providing test reports from laboratories accredited to carry out the relevant tests and/or calculations and appropriate documentation of production methods and quality controls.

PRODUCT TYPE	MAXIMUM VOC CONTENT (G/LITRE)
Latex Primer for galvanised iron and zincalume	60
Exterior latex undercoat	55
Interior latex undercoat	65
Interior gloss	75
Interior semi-gloss	16
Exterior gloss	75

- VOC limits on architectural coatings covered. Allowable levels include water content in the formulation.
- For solvent-based coatings the paint shall not contain VOCs in excess of 200g/litre. For recycled paints the VOC level (averaged across batches) in the paint must not exceed 100g/litre.
- The VOC content of the paint shall either be calculated from the VOC data for each of the raw materials or, experimentally by ASTM D3960, as qualified the Australian Environmental Labelling Association, Inc. Standard No: AELA 23-2005 'Australian Voluntary Environmental Labelling Standard Architectural and Protective Coatings'.
- Where the raw material is a mixture of compounds, some of which contain VOCs, the VOC content of the mixture may in turn be calculated from the VOC content of the individual components. Where this is not known, it must be determined by the methodology detailed in AELA 23-2005.

5.4.2 PREPARATION OF SURFACES

- Preparation: all surfaces shall be prepared by thorough cleaning of all dirt, grease, scale, welding flux and corrosion in accordance with AS/NZS 2312.
- For galvanised surfaces that have been subsequently cut or welded, prime the affected area with organic zinc rich coating for protection of steel to AS/NZS2312, two pack or accepted alternative.
- Prime/undercoat all surfaces. Galvanised and non-ferrous surfaces must be etch primed.

5.4.3 APPLICATION

- 1 First coat: apply the first coat immediately after the substrate preparation and before contamination of the substrate can occur.
- 2 Apply all paint systems in strict accordance with manufacturer's recommendations.
- 3 Finish: each coat of paint or finish shall be uniform in colour, gloss, thickness and texture and free of runs, sags or blisters.

5.4.4 FINISHING

- 1 Concrete Bases: 2 coats synthetic emulsion paint
- 2 Equipment Panels/Switchboards: 2 final coats full gloss enamel
- 3 Equipment: Factory spray application
- 4 Other surfaces: 2 final coats full gloss enamel

5.5 COLOURS

Paint colours nominated are to be in accordance with AS 2700 and AS 1345.

5.6 IDENTIFICATION AND LABELLING

5.6.1 GENERAL

- 1 Mark all systems and equipment including wiring, piping, control panels, switchboards, ductwork, equipment cabinets, conduits, controls, gauges, valves and fittings so that they are readily identifiable.
- 2 Equipment requiring special maintenance procedures or presenting occupational health and safety hazards: provide permanent labels complying with statutory requirements.
- 3 Locate labels so that they are easily seen and either attached to, below or next to item referred.
- 4 Labelling to match installation documentation schedules.
- 5 For systems containing hazardous materials provide labelling in accordance with AS 1318.
- 6 Where a pipe or duct passes through a wall, slab or partition provide labelling each side.
- 7 Equipment and Service labels shall be in accordance with the Legend of Symbols nomenclature for the respective service. Provide appropriate Emergency and Operating Instructions to every equipment and system.
- 8 Provide text and simple diagrams where practicable. Locate at a fixed point and point of operation.
- 9 Affix hazardous equipment with warning signs.
- 10 Labels shall be permanent.
- 11 Type: select from the following:
 - a Engraved and red filled lettering on stainless steel / brass plate or plastic (traffolyte), minimum 1mm thick and mechanically fixed.
 - b Stencil with red or white lettering or of a contrasting colour with the background.
 - c Engraved two-colour laminated plastic mechanically fixed.
 - d Adhesive labelling.

e Painted.

12 Adhesive labels and laminated plastic labels shall not be used where exposed to the environment.

13 Painted labels shall be in the same finish paint type as the installation.

5.6.2 LABELLING REQUIREMENTS

ITEMS	REQUIREMENTS
Pipework (Fire)	Labelling and lettering to AS 1345 and AS 1318
Conduits	Labelling and lettering to AS 1345
Exposed Conduits	Labelling and lettering to AS 1345
Valves	Labels shall be colour coded laminated labels indicating service, function and normal position, and attached to hand wheels or spindles using a vandal resistant permanent chain/ring
FDCIE'S and EICIE Panel (General)	— Main FDCIE Panels: Words Stating: "FIRE PANEL" on the front of the panel / doors in letters no less than 50mm high in contrasting colour to background colour scheme. — Main EICIE Panels: Words Stating: "EWIS PANEL" on the front of the panel / doors in letters no less than 50mm high in contrasting colour to background colour scheme. — Outgoing electrical functional units: 8mm. — Automatic controls dry fire / electrical equipment and instruments: 5mm. — Rear covers of fire panels: 5mm. — Components inside dry fire / electrical enclosures and control panels: 3.5mm — Locate labels so that they are easily seen from normal access on front of the item being marked. Do not install labels on components normally removed or replaced.
All equipment such as pumps, including controls, gauges and fittings to that they are readily identifiable	— Label in accordance with the specification abbreviations, symbols and acronyms scheduled or otherwise approved. — Locate labels so that they are easily seen from normal access adjacent to the item being marked. Do not install labels on components normally removed or replaced. — Major equipment nameplates: 40mm. — Minor equipment nameplates: 20mm. — Danger, warning and caution notices: 10mm for heading, 5mm for text. — Warning notices: 7mm. — Minor lettering: 3mm. — Lettering Style: Helvetica Medium. — Fixing: Use mechanical fixings. Do not penetrate isolation vapour barriers. — Equipment requiring special maintenance procedures or presenting occupational health and safety hazards: provide permanent labels complying with statutory requirements. — Equipment requiring consumables including replacement belts, oils, filters and strainers: provide labels indicating consumable component details and quantities and corrective maintenance trigger point such as filter pressure drop.
Fire services In ceiling identifications	1 Provide identification to ceiling tile or access panel for servicing to test drain valves, flow switches, concealed detectors or similar.

ITEMS	REQUIREMENTS
	2 Use colour markers of 'black' for general servicing and 'red' for Code required identification e.g. concealed detectors.

6 INSPECTION, TESTING, COMMISSIONING, HANDOVER AND DEFECTS LIABILITY PERIOD

6.1 GENERAL

- 1 This section sets out the engineering and quality assurance requirements for the completion of the installation, inspection regime, testing, commissioning, handover and defects liability period.

6.1.1 DESIGN

Complete the detail engineering of the testing commissioning and handover of the works to the specified requirements.

Comply with applicable authority design guidelines and codes of practice and requirements.

As a guide, commissioning guidelines as detailed by CIBSE Commissioning Codes shall be used as a minimum to undertake the commissioning of services listed in this section.

6.1.2 STANDARDS & GUIDELINES

AS 1851 Maintenance of fire protection systems and equipment

Also refer to Design and Performance Criteria section

6.2 PRACTICAL COMPLETION REQUIREMENTS

- 1 Notwithstanding any other provisions of the Contract with regard to the granting of Practical Completion, the following requirements, listed in order of importance / priority, shall be completed and completion submissions made prior the date of Practical Completion.
- 2 Minimum submissions required for application for Occupancy Permit
- 3 Submission of signed off testing and commissioning schedules to confirm that Testing and Commissioning has been finalised and that all plants and systems operate in a safe, stable and automatic manner under all conditions of full and partial load - full commissioning data.
- 4 Submission of Authority approvals including Statutory Authority approvals and Certification of Compliance with the NCC
- 5 Submission of Certification that works comply with the Contract Documents
- 6 Operating Instruction and Maintenance manual sections covering all Safety Services as required by regulations
- 7 Other submissions:
- 8 Submission of correctly and completely executed ITPs including test results completed in accordance with the Contractor Quality Plan and the satisfaction of the Consultant Engineer
- 9 Adequate training and instruction of the Principal's representatives in safe operation of the plant
- 10 Official Equipment manufacturer's acceptance certification for the installation
- 11 Completed Operating and Maintenance manuals and 'as built' drawings

12 The Defects Liability period may be extended in the event:

- a Specific systems fail to achieve the desired outcomes after two attempts are made to rectify the drawback. These specific systems include those that are fire and life safety or business / mission critical to the Principal.
- b If there have been more than three (3) attempts to rectify the defect, in which instance a further 12 months of monitoring will be required after the defect is rectified to monitor that the desired outcomes are achieved without further shortfalls.

6.3 MANAGEMENT OF INSPECTION, TESTING, COMMISSIONING, HANDOVER AND DEFECTS LIABILITY PERIOD COMMITMENTS

6.3.1 GENERAL

- 1 Manage the whole process so as to;
- 2 Ensure complete execution of Project Specific Quality Plan and specifically Installation and Commissioning ITPs are completed progressively from commencement of the project.
- 3 Test and verify the operation of individual items of plant and equipment, sub-systems, systems and the overall installation including the interfaces with other building services systems forming part of the project.
- 4 Create and record accurately and verify all “As Built “drawings progressively prior to services being concealed by construction of ceilings, wall cavities and shafts.
- 5 Ensure completed Operating and Maintenance manuals inclusive of comprehensive system descriptions, operating modes and associated control functionality, tailored and project specific manufacturers literature.
- 6 Responsive defects rectification process during the construction phase as well as post Practical Completion during Defects liability Period.
- 7 Ensure Responsive approach towards maintenance and operational fine tuning during the Defects Liability Period

6.3.2 SEQUENCE OF ACTIVITIES

Generally, follow the sequence of activities listed below. Modify to suit specific requirements of the project however no reduction of scope is allowed.

- 1 Submission of Project specific Project Quality Plan including generic and project Specific ITPs within the first quarter of the project programme after site possession.
- 2 Ongoing inspection of installation works by relevant Quality Manager and/or his nominated staff for meeting project quality requirements including preparing Inspection reports (listing all observed installation defects) not longer than fortnightly and also checking that all Installation ITPs are signed off.
- 3 Initiate Safety in Design process and documentation where applicable
- 4 Evidence of adequate commissioning programme allowance in construction programme.
- 5 Production and submission of commissioning Inspection and Test Plan (ITP) 12 weeks after site possession
- 6 Amendment to ITP as necessary and re-submission at least 1-2 weeks after the original submission.
- 7 Submission of controls manufacturer’s functional descriptions elaborating on the methodology proposed to implement the requirements of the control strategies. The manufacturer’s controls functional descriptions shall be prepared and submitted at least 12 weeks prior to start of installation and commissioning of controls systems.

- 8 Nomination of those to be involved in commissioning including individuals with responsibility for management, engineering and field testing and commissioning.
- 9 Liaison with and briefing of Authorities to ensure that the connection to their assets and commissioning procedures and resultant data provided for their approval is consistent with their requirements and records.
- 10 Preparation of periodic (minimum monthly) progress reports on testing and commissioning.
- 11 Early and active engagement of Independent Commissioning Agent ICA where applicable.
- 12 Early (within the first quarter of the project construction programme) submissions of Operating and Maintenance Manual (for Safety Services and, for preliminary draft of the overall manual). Confirmation of acceptance of proposed Electronic Operating and Maintenance Manual software/platform.
- 13 Submission of detailed commissioning procedures for review by the Consultant Engineer. Submission shall be made early, at least 12 weeks prior to the commencement of commissioning when the project construction period greater than 12 months, or 6 weeks when the project construction period is less than 12 months duration.

EXPECTED PROJECT DURATION (UP TO PC)	SUBMISSION REQUIREMENT	RE-SUBMISSION REQUIREMENT.
8 weeks	Prior to Week 1 or Prior to commitment of work whichever the sooner.	Within 3 days.
12 weeks	Prior to Week 1 or Prior to commitment of work whichever the sooner.	Within 5 days
16 weeks	Prior to Week 2 or Prior to commitment of work whichever the sooner.	Within 5 days
20 weeks	Etc.	Within 5 days
24 weeks		Within 10 days.
9 months		Within 10 days.
12 months +		Within 10 days.

- 14 Rectification of all defects likely to interfere with testing and commissioning.
- 15 Commencement of “As Built” drawing production
- 16 Carryout Preliminary commissioning including pre-commissioning procedures until achievement of correct operation and performance. Providing statement that the pre-commissioning activities for each system are completed.
- 17 Testing and commissioning of major plant and equipment, complete systems, interfaces between systems and finally all systems which interact together under normal or emergency conditions.
- 18 Integrated testing and commissioning of multi systems (including multi-disciplinary systems) and their associated interfaces
- 19 Inclusion of all completed Commissioning ITPs in the Operating and Maintenance Manual.
- 20 Rectification of any further defects and deficiencies found during testing and commissioning.
- 21 Provide notification of completion of remedial work in sufficient time to permit re-inspection as necessary before the intended date for re-testing.

- 22 Repetition of sequence of activities as necessary until works comply with acceptance criteria as set out in the ITPs.
- 23 Final handover of Final Operating and Maintenance Manuals
- 24 Certification that works comply with the Contract Documents and with the NCC and Statutory Authorities and are ready for granting of Practical Completion.

6.3.3 SUBMISSIONS

Required submissions include but not limited to the following;

- 1 Installation and commissioning Inspection and Test Plans ITPs. 4 Copies of ITPs to be supplied, 3 bound into Installation Manuals and 1 bound separately
- 2 Evidence of periodic (weekly) Quality Inspection/defects reports
- 3 Testing and commissioning procedures including integrated commissioning procedures.
- 4 Evidence of Safety In Design process during the construction phase is completed where applicable
- 5 Evidence of currency of calibration of equipment to be used in commissioning procedures.
- 6 Records of all pre-commissioning checks and final commissioning data. All records/data to be certified as being conducted in accordance with agreed and approved procedures.
- 7 Manufacturers' certificates for all proprietary items, to confirm that the installation complies with their installation and maintenance requirements prior to start-up of equipment. Certificates shall also be provided at commencement of equipment installation to confirm their requirements with respect to installation are being provided for.
- 8 Manufacturers' test certificates of all capacity/performance verification checks carried out at factory prior to dispatch to site.
- 9 Accurate and verified (by QA manager) "As Built" drawings
- 10 Complete Operating and Maintenance Manuals (Electronic system/package and, unless instructed otherwise, hard copies)
- 11 Warranties
- 12 Maintenance Plan and Schedules following acceptance of technical submission and installation minimum 4-8 weeks prior to Practical Completion with priority with respect to timely completion given to plan and schedules covering Safety Services.

6.3.4 INSPECTION AND TEST PLANS

- 1 Inspection and test plans are to be prepared specifically for the project but may be based on or customised from generic ITPs. All commissioning result forms shall be completed, including the following:

Tested By: Please Print	Witnessed By:
Signed:	Signed:
Date:	Date:
Test Device used:	
Serial Number:	
Date of Calibration:	
Method of Testing:	

- 2 List acceptance criteria for each element and sub-element of the installation in ITPs. Acceptance criteria to include:
 - a Specification details in regard to materials, construction methods, physical requirements, performance and operational requirements.
 - b Schedule of Technical Data in respect of make, model and performance details of equipment.
 - c Shop drawings.
 - d Control logic and diagrams.
 - e Authority inspection and testing requirements.
 - f References to specific clauses of this Specification for acceptance criteria

6.3.5 COMMISSIONING INSTRUMENTATION

- 1 All instrumentation used in the commissioning of the installation shall be managed in accordance with the requirements of AS 3912 or NEBB.
- 2 Reference instruments are to be NATA certified.
- 3 All commissioning results to include instrument calibration documentation.

6.3.6 COMMISSIONING PERSONNEL

- 1 Provide only qualified registered / licensed personnel specific to the trades as required by local regulations, and where regulatory requirements do not apply, personnel certified by appropriate industry bodies.
- 2 Submit a schedule of key commissioning personnel together with their qualifications prior to undertaking commissioning activities on site.

6.4 TESTING & COMMISSIONING

6.4.1 PRE-ENERGISATION TESTS

Prior to energisation of any system with electrical connections, conduct the following tests:

- 1 Tightness of screwed and bolted connections.
- 2 Physical integrity.

- 3 Correct phasing. Check phase rotation is consistent throughout the project and is the same as the supply.
- 4 Motor rotation.
- 5 Insulation resistance tests.
- 6 Test operation of all trips, interlocks, motor driven devices, contactors and control circuits and devices by instigating or simulating inputs.

6.4.2 TESTS

- 1 Comply with Australian Standards for testing and commissioning comprising:
 - a Water systems – static, flow and hydrostatic pressures.
 - b Automatic fire detection alarms.
 - c Occupant warning noise level verification.
 - d Electrical systems – power and controls.
 - e Gauges and instrumentation.

6.4.3 COMMISSIONING

Operate the installation to prove the performance, capacities and ability to provide the required service in accordance with the applicable fire systems Australian Standards.

Undertake commissioning in 4 stages:

- 1 Individual component functionality testing including verification of defects free installation
- 2 Major items of plant and equipment and components.
- 3 Systems and sub-systems.
- 4 Inter-system operation and interfaces.

Commissioning to include:

- 5 Performance of individual plant items and components.
- 6 Operating sequences, interlocks and safeties.
- 7 Final controls calibration.
- 8 System operation under all operating modes and under all conditions of load.
- 9 Inter-system operation and correct interfacing connections under all operating conditions and under simulated fire conditions.
- 10 Noise and vibration tests.
- 11 Thermos can survey of switchboards.
- 12 Environmental audit as required by Authorities.
- 13 Rectification and correction of any defects and deficiencies.
- 14 Continue commissioning until achievement of correct performance and operation.
- 15 Electrical testing and verification to AS/NZS 3000.
- 16 Cleaning of plant room, switchboards and all equipment and devices.

6.5 FINAL CONTRACT AND HANDOVER DOCUMENTATION

6.5.1 GENERAL REQUIREMENTS

- Documentation is required in electronic and hard copy format. Electronic documentation shall be supplied on the latest Microsoft, AutoCAD and Revit Software versions at the time of Practical Completion.
- Scope of the documentation and format requirements comprise:

DOCUMENT	FORMAT	SETS
Operating and Maintenance Manuals, including Final commissioning data	Electronic installed on BAS File Server	1
	USB FLASH DRIVE – Microsoft Office Latest Version	3
	Hardcopy	3
As Built Drawings	USB FLASH DRIVE – AutoCAD	3
	Hardcopy – Full Size with hanging strips	1
	A3 Size	2
Control Software	USB FLASH DRIVE	1
OH&S Safety Signage	Permanent	1

6.5.2 AS BUILT DRAWINGS

- All As Built/As Installed documents submitted shall have a stamp with similar words to the following and be signed prior to submission for review.

AS INSTALLED DRAWING
We certify that this drawing is an accurate record of the work installed by our company under the terms of our contract. Contractor's Name Contractor's Address Signed Date Printed

- Progressively record changes to form a record of work as installed.
- Prepare and submit for approval as-built drawings covering all aspects of the work.
- Show dimensions, types and locations of equipment, cables, piping, ductwork pits and markers in relation to permanent site features and other underground services.
- Show the 'as-installed' locations of building elements, plant and equipment with particular emphasis on items requiring maintenance or clearing.
- Show off-the-grid dimensions where applicable. Include relationship to building structure street features and other services and changes made during commissioning.

- 7 Show inverts of drainage and relative locations of gully and boundary traps
- 8 Provide As Built documentation as per Schedule of Required As Builts:

6.5.3 CERTIFICATION

- 1 Provide certification for all systems designed, installed and maintained as required under the Building Act 1975.

6.5.4 OPERATING AND MAINTENANCE MANUALS

Provide comprehensive Operating and Maintenance documentation to cover all installations in order to provide a detailed understanding of the plant and its operation, an aid for training of operators, a reference for fault diagnosis and a framework for preventative and breakdown maintenance.

Submit manuals in two stages. Stage 1 include all of the sections (generally sections 1-5) of manuals excluding final testing and commissioning records and As Built drawings. This manual shall be submitted well in advance of the Practical Completion and should be signed off by the Consultant Engineer at least 4-6 weeks before programme Practical Completion Date.

- 1 First draft manuals Stage 1: Submit a first draft manual 8-10 weeks before the date for practical completion for review.
- 2 Second drafts Stage 1: Following receipt of comment on the first draft and not later than 2 weeks after submitting a completed second draft for review. Incorporate comments.
- 3 Third Draft Stage 1 & 2: Include all remaining sections and submit 2-3 weeks before Practical Completion. Incorporate all changes and resubmit until all comments are addressed to the satisfaction of the Consultant Engineer.
- 4 Final copies Stage 1&2: Submit 3 sets of final volumes within 2 weeks after practical completion. Incorporate feedback from review and from training of Principal's staff, including preparation and insertion of additional data. Include a section containing commissioning test reports.
- 5 Revisions: Prior to final completion, submit 3 sections sets of loose-leaf amendments for insertion in the manuals, incorporating feedback from the maintenance period.
- 6 Warning and Cautions: Include to emphasise conditions hazardous to personnel or equipment, giving instructions to avoid the hazard. Format to be:
 - a **WARNING:** An examining or testing procedure or practice which must be observed or risk loss of life or injury to personnel.
 - b **CAUTION:** An examining or testing procedure or practice which must be followed or risk damage to equipment.
- 7 The manual shall be neatly prepared and bound in one, or a series of, vinyl hard-back folder with lettering on the front and the spine including the following information:

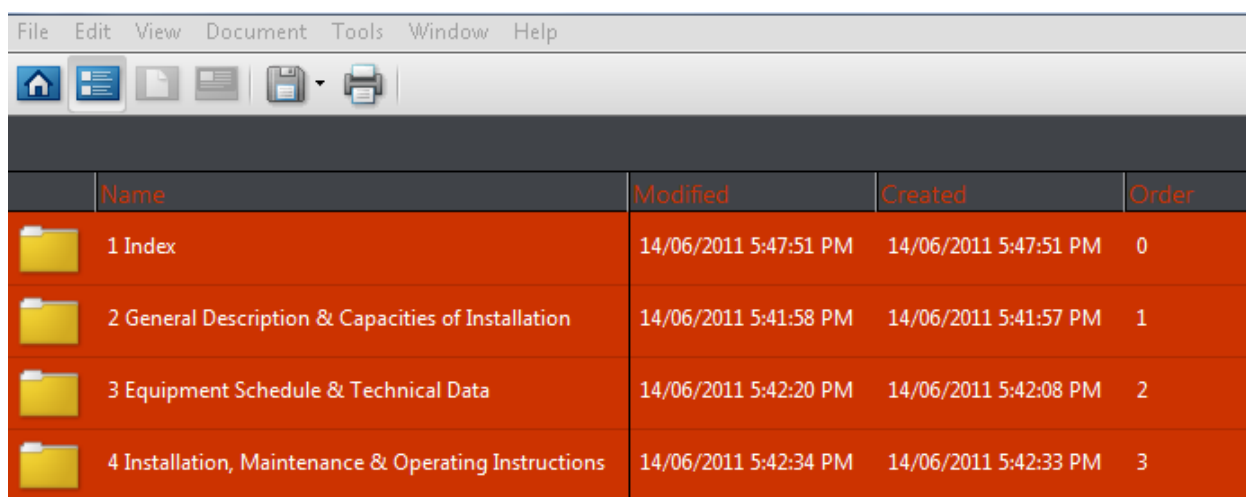
<u>OPERATION AND MAINTENANCE MANUAL</u>	<u>VOLUME NUMBER</u>
<u>PROJECT NAME</u>	<u>DATE OF ISSUE</u>
<u>FIRE PROTECTION SERVICES</u>	<u>CONTRACTOR</u>
<u>CONSULTANT ENGINEERS – WSP</u>	<u>SUBCONTRACTOR</u>

- 8 Pagination: Number pages consecutively. Double sided on A4 size (minimum 50%) recycled white (minimum CIE whiteness of 140) paper, with reinforced binder perforations.
- 9 Ring size: 50mm maximum, with compressor bars.

- 10 Dividers: Durable divider for each separate element, with typed description of system and major equipment components. Clearly print short titles under laminated plastic tabs.
- 11 Drawings: Fold illustrative drawings to A4 size and accommodate them in plastic pockets in the binder. Title block to be visible without removal of drawing
- 12 All aspects of the style and quality of the manual, including folders and contents shall be submitted for approval prior to completion.
- 13 After the draft copy has been approved, supply 3 copies of the manuals for issue to the Principal and the Consultant Engineer.
- 14 Allow for loading the manual onto the Building BAS systems (where available) and arrange to provide the manual including commissioning results.

6.5.4.1 ELECTRONIC/DIGITAL FORMAT

- 1 Documents shall be stored in USB Flash Drive. Provide labelling to the storage media.
- 2 Electronic Storage device shall not be “copy protected”.
- 3 Manual shall be contained within a **single searchable PDF** accessible via Adobe or similar PDF reader. PDF shall be indexed, divided into folders and titled. Each folder shall contain relevant electronic/digital format documentation in line with the contents specified below. Refer below of a screen print of Adobe Reader for a typical arrangement.



	Name	Modified	Created	Order
	1 Index	14/06/2011 5:47:51 PM	14/06/2011 5:47:51 PM	0
	2 General Description & Capacities of Installation	14/06/2011 5:41:58 PM	14/06/2011 5:41:57 PM	1
	3 Equipment Schedule & Technical Data	14/06/2011 5:42:20 PM	14/06/2011 5:42:08 PM	2
	4 Installation, Maintenance & Operating Instructions	14/06/2011 5:42:34 PM	14/06/2011 5:42:33 PM	3

“Screen Shot” for Reference

- 4 Avoid submitting manually scanned Manufacturers’ data, where possible include only searchable PDF documentation.
- 5 Include only relevant diagrams, equipment data in clear concise English. Avoid jargon and esoteric pseudonyms.
- 6 Provide each document included in the PDF with the following naming convention:
- 7 Section “X” – “Document Description”.pdf
- 8 Provide clear scanned copies of Statutory Certificates of Compliance
 - a Accessible Dwg format CAD drawings of the as-built documentation shall be contained within the PDF.

6.5.4.2 INSTALLATION MANUAL FORMAT

- 1 Section 1 – Index
 - a Index all sub-divisions of each section including lists of drawings, equipment, etc. for quick reference.

- b** Include index in every volume of the manuals.

2 Section 2 - General Description and Capacities of Installations:

- a** Divide into sub-sections for general items and for each individual system as appropriate and include full details
- b** Include a directory: Names, addresses and telephone, email and facsimile numbers of Principal, Consultant, Sub-Consultants, Contractor and names of responsible parties.
- c** Include a scope of works: Statement of scope of services and interfacing with other Contracts.
- d** Include an installation description: General description of installation.
- e** Systems description: Technical description of the systems installed, written to ensure that the Principal's staff fully understand the scope and facilities provided. Identify function, normal operating characteristics and limiting conditions. Include schematic diagrams.
- f** System performance: Technical description of the modes of operation of the systems installed.

3 Section 3 – Equipment:

- a** Sub-divide as for Section 2 and list all major items of equipment installed complete with manufacturer's name, agent's name, contact details, model and/or type No., serial No., size, design ratings and including all relevant data necessary for re-ordering or replacing the equipment or components of same. As far as practicable, all equipment should be broken down to individually identifiable items.
- b** Provide manufacturers' technical literature for equipment installed, assembled specifically for the project, excluding irrelevant matter. Mark each product data sheet to clearly identify specific products and component parts used in the installation and data applicable to the installation. Do not include advertising literature. Cross out any section of literature which is not applicable to the project.

4 Section 4 – Installation, Maintenance and Operating Instructions:

- a** Sub-divide as for Section 2 and include manufacturer's installation, maintenance and operating instructions for each item of equipment.
- b** Emergency maintenance procedures, including telephone numbers for emergency services and after hour's contact for suppliers and Contractors and procedures for fault finding.
- c** Manufacturer's technical literature as appropriate. Register with manufacturer as necessary. Retain copies delivered with equipment.
- d** Detailed recommendations for preventative maintenance frequency and procedures which should be adopted by the Principal to ensure the most efficient operation of the systems installed. Include inspection, testing and maintenance programme in tabular form showing frequency and level of routine checks for each item.
- e** Safe trouble shooting, disassembly, repair and reassembly, cleaning, alignment and adjustment, balancing and checking procedures. Provide logical step-by-step sequence of instructions for each procedure. Include calibration and recommissioning of controls.
- f** Schedule of spares recommended to be held on site, being those items subject to wear or deterioration and which may involve the Principal in extended deliveries when replacements are required. Include complete nomenclature and model numbers, and local sources of supply. Include sectionalised diagrams of machines identifying component parts.
- g** Schedule of normal consumable items, local sources of supply and expected replacement intervals up to a running time of 40,000 hours. Include lubricant and lubrication schedules for equipment where applicable.
- h** Add to the manufacturer's technical literature, for assembly, operating, maintenance and disassembly, safety information for all plant, systems or structures installed within the project or for existing plant, systems or

structures where assembly, operating, maintenance and disassembly are altered or amended by the installation of plant, systems or structures within the project.

- i The added safety information shall be included at relevant sections of the manufacturer's literature and not as a separate part and shall be made clear and obvious to the user of the manual.

5 Section 5 – Plant Operating Instructions:

- a Sub-divide as for Section 2 and provide a complete description and correct sequence of all actions necessary to start-up, operate and shut-down all plant including procedures for seasonal changeovers where applicable. These instructions shall include full information on such items as normal and abnormal gauge readings, instrument settings and control points, differentials, time delays, oil levels, water temperatures, and all similar relevant variable and adjustable items, to permit checking and adjustments where practicable and identification of hazardous conditions or malfunction of plant. In addition, the instructions shall include information on the immediate action to be taken in the event of hazardous conditions arising.
- b Add to the manufacturer's technical literature, for assembly, operating, maintenance and disassembly, safety information for all plant, systems or structures installed within the project or for existing plant, systems or structures where assembly, operating, maintenance and disassembly are altered or amended by the installation of plant, systems or structures within the project.
- c The added safety information shall be included at relevant sections of the manufacturer's literature and not as a separate part and shall be made clear and obvious to the user of the manual.

6 Section 6 – Performance Test Results:

- a This section shall be sub-divided as for Section 2 and shall include all performance test results as outlined in previous Clauses.

7 Section 7 – Certificates & Warranties

- a Installers statements for all Fire protection systems installed for this project
- b Formal Certificate by Independent Fire certifier for all Fire protection systems installed
- c Statutory Certificates of Compliance for:
 - d Electrical work.
 - e Plumbing drainage or gas-fitting work.
 - f Refrigeration or air conditioning and ventilation work.
 - g Copies of manufacturers' warranties.
 - h Certificates from authorities and utilities.
 - i Production certification.
 - j Copies of test certificates for the installation and equipment used in the installation.

8 Section 8 – Drawings

- a Drawings and technical data: As necessary for the efficient operation and maintenance of the installation.

6.5.5 WARRANTIES

- 1 Warranty periods are deemed to end at expiry of defects liability period unless specified otherwise.
- 2 Name the Principal as warrantee. Register with manufacturers as necessary. Retain copies delivered with components and equipment.

- 3 Commence warranty periods at practical completion or at acceptance of installation, if acceptance is not concurrent with Practical Completion.
- 4 If installation is not being manufactured and product warranty is conditional on the manufacturer's approval of the Contractor, submit the manufacturer's written approval of the installing firm.

6.5.6 REQUIRED OPERATING INSTRUCTIONS

- 1 Provide operating instructions and schematic diagrams mounted alongside equipment to meet statutory requirements, e.g. Fire Brigade. Diagrams to be colour-coded, colour-fast, laminated or engraved and mounted.

6.5.7 TRAINING OF PRINCIPAL'S NOMINATED PERSONNEL

- 1 The service of a fully knowledgeable Commissioning Agent shall be required for a period of 1 full working weeks to instruct the Owner's nominated representatives in all details of the plant operation.
- 2 Conduct training at agreed time, at system or equipment location.
- 3 Immediately after Practical Completion, explain and demonstrate to the building users or nominated representative, the purpose, function and maintenance of the installations.
- 4 Use qualified personnel who are knowledgeable about the installations.
- 5 Prior to Practical Completion, explain and demonstrate to the Principal's nominated personnel the purpose, function, operating and maintenance of the installations.
- 6 Use items and procedures listed in the Operation and Maintenance manuals as the basis for detailed instruction of Principal's nominated personnel.
- 7 Conduct training at agreed time, at each system of equipment location.
- 8 Upon completion, submit certificates of training, signed by trainer and trainees.
- 9 Provide qualified manufacturer's training representatives who are knowledgeable about the installations
- 10 During the warranty period, provide technical assistance and advice to the Principal's nominated personnel regarding the operation and maintenance of the plant.
- 11 For equipment requiring seasonal operation, demonstrate during the appropriate season and within 6 months of Practical Completion.
- 12 Include copy of training material within the O&M Manual

6.5.8 SPARE PARTS SCHEDULE

At least 8 weeks before the date for Practical Completion, submit a schedule of spare parts necessary for maintenance of the installation. State against each item the recommended quantity, and the manufacturer's current price, including for

- 1 Packaging and delivery to site;
- 2 Checking receipt, marking and numbering in accordance with the spare parts schedule;
- 3 Referencing equipment schedules in the operation and maintenance manual; and
- 4 Painting, greasing and packing to prevent deterioration during storage.

6.6 POST PRACTICAL COMPLETION AND DEFECTS LIABILITY PERIOD

6.6.1 WARRANTY & DEFECTS LIABILITY

Warrant the whole of the Contract Works, for the Defects Liability Period (DLP), against defective workmanship and materials and against non-compliance of equipment and/or complete system with specified performance and operation.

The Defects Liability Period shall continue for a period of 12 months after the date of issue of Practical Completion and during this period, be responsible for making good on a timely manner, defects arising from the defective design materials, premature component failures, or workmanship or from any act of the Contractor that may develop in the work.

Perform maintenance during the warranty period to maintain warranty conditions. Maintenance shall comply with the current regulatory requirements.

In the event of inclusion of equipment normally covered by a lesser time warranty, or whose warranty would otherwise normally expire during the warranty period, allow for and include the costs of extending such warranty to that covering up to the end of the Defect Liability Period.

Where warranted work or equipment has failed, the warranty shall recommence on the date of acceptance of the rectified work for an additional period of time, extended by the period during which the Principal was without the use of the work or equipment. For rectified work involving new replaced equipment, the warranty for the new component shall be from date of replacement for the 12 months or longer as required by this specification

6.6.2 GENERAL DLP MAINTENANCE SCOPE

Maintenance services shall include:

- 1 Preparation of a maintenance management plan which includes a detailed maintenance programme, equipment schedules, statutory and preventative maintenance schedules, OH&S risk assessment, inspection and testing schedules, breakdown rectification management and reporting systems. Submit maintenance plan and schedules prior to Practical Completion
- 2 Preventative and corrective maintenance and testing services to assure specified system reliability and availability.
- 3 Statutory maintenance and testing required by the Building Regulations and other authorities.
- 4 Breakdown and system fault responses. Attend site, ascertain cause of alarm and rectify where possible. Make system alterations at own cost to prevent future system faults. Pay any Authority charges resulting from callout on system faults.
- 5 Statutory regular system fire and life safety system control and performance verification test to be carried out after hours.
- 6 Comply with Building and Occupational Health and Safety Regulations.
- 7 Comply with manufacturer's recommended maintenance programme and schedules.
- 8 Monthly maintenance and servicing in accordance with maintenance schedules recommended by equipment manufacturers.
- 9 Rectification of defects at Contractors cost.
- 10 Replacement of consumables at Contractors cost during the maintenance period.
- 11 Submission of monthly reports to Principal and Consultant Engineer on maintenance activities performed.
- 12 Provision and filling out logbook which must be kept on site, detailing all visits.

- 13 Service personnel to be available for call out on a 24 hour per day, 7 day per week basis. Response time for emergency and corrective maintenance must not exceed 2 hours.
- 14 Operation and verification in conjunction with other services trades, of all fire and life safety systems.

6.6.3 FINAL COMPLETION

Final Completion will not be granted until maintenance records are complete and signed off and until all defects have been rectified and inspected.

The Maintenance and Defects Liability Period may be extended unless maintenance and servicing visits have been made and reported as specified.

6.6.4 SUBMISSIONS

- 1 Provide;
 - a Maintenance management plan.
 - b Monthly Maintenance reports.
 - c Completed maintenance records

Mandatory maintenance records to include monthly maintenance and service reports, including water treatment reports, reports on all defects and their rectification, certification of the correct operation of essential services, a maintenance logbook and any other records which must be kept in order to comply with legislative or OH&S requirements.

- 2 Monthly Maintenance and Service Reports to include:
 - a Check list of all items serviced and inspected.
 - b Description of maintenance performed.
 - c Notes on supplementary maintenance required.
 - d Comprehensive water treatment report.
- 3 Maintenance Logbook to include:
 - a Schedule of all equipment.
 - b Schedule of all control sensors and control set point values.
 - c Register of maintenance visits and cross reference to service reports.
 - d Register of work performed under headings of preventive maintenance, corrective and emergency maintenance and defects rectification.

The Maintenance Logbook is to be kept in a locked cabinet provided for the purpose in an agreed location on site.

At the conclusion of each maintenance visit the logbook is to be duly completed and signed off by an authorised representative of the Principal /Building Owner and service records are to be forwarded to the Principal /Building Owner or his representative within 3 days of each visit.

6.6.5 DLP PREVENTATIVE MAINTENANCE AND TESTING

6.6.5.1 GENERAL

- 1 Carry out routine maintenance and servicing commencing recommended by the manufacturer of each piece of equipment at the recommended intervals from the date of Practical Completion.
- 2 Maintenance and servicing shall include:

- a Maintenance, testing and servicing in accordance with maintenance schedules required by the Certificate of Occupancy/Classification.
 - b Periodic maintenance, testing, servicing and inspections in accordance with regulatory requirements and relevant Australian Standards.
 - c Submission of monthly reports on maintenance activities performed.
- 3 Ensure adequately qualified and experienced service personnel carry out the works.
- 4 Use approved and calibrated instruments for any required testing or servicing.
- 5 Report to the designated representative of the building on arriving at and before leaving the building/site.
- 6 Provide a service which is available for call out on a 24 hour per day, 7 day per week basis. Response time for emergency and corrective maintenance must not exceed 2 hours.

6.6.5.2 REFERENCED DOCUMENTS

Details of documents referenced and applicable to this section of the specification are as follows:

Australian / New Zealand Standards

STANDARD	TITLE
AS 1851	Maintenance of fire protection systems and equipment

6.6.5.3 MAINTENANCE PROCEDURES

- 1 Comply with standards.
- 2 Allow to conduct, liaise and manage any co-ordinated testing with other services contracts where procedures for integrated systems testing is required by Codes or as part of approval at time of initial handover.

6.6.5.4 RECORDS

- 1 Record details of all site visits whether of routine or emergency nature.
- 2 Replace logbooks as they become due.
- 3 Maintain logbooks in accordance with legislative requirements and approvals pertaining to the building at the time of handover.
- 4 Provide proof of attendance and completion of all routine servicing visits.

6.6.5.5 SYSTEM FAULTS

- 1 Attend site.
- 2 Ascertain cause of alarm and rectify where possible.
- 3 Make system alterations at own cost to prevent future system faults.
- 4 Pay Authority charges resulting from callout on system faults.

6.6.5.6 ONGOING MAINTENANCE AND SERVICING

- 1 Submit a proposal to continue with routine maintenance and servicing within 8 weeks prior to the end of the Defects Warranty and Liability period.
- 2 Provide details of the procedures to be carried out.

- 3 List additional levels of service other than general which relate to equipment/system overhaul and/or other long term special levels of service.
- 4 Identify callout rates for emergencies, false alarms, faults or acts of vandalism.

7 EQUIPMENT AND SYSTEMS

7.1 FIRE HYDRANT AND HOSE REEL SYSTEMS

7.1.1 HYDRANT MILLCOCK

- 1 Provide SHIELD Global Hydrant Landing Valves to match CQU campus existing hydrants.
- 2 Provide connections compatible with equipment used by Fire Authority.
- 3 Include brass cap chained to valve body.
- 4 Locate at correct height above finished ground/floor access level.
- 5 Maintain clearances to hand-wheels and connections.

7.1.2 FIRE HOSE REEL

- 1 Length of reel minimum 36m.
- 2 Nozzle of brass construction (plastic not acceptable).
- 3 Fit with an approved RPZD backflow prevention device where required.
- 4 Install extended hose guidance arm to ensure protrusion outside the recess or cabinet when reel in use.

7.1.3 PIPING AND FITTINGS – HYDRANT

- 1 Comply with piping manufacturing Codes:
 - a AS 1074 for steel piping, medium weight grade as a minimum. Light wall tubing not permitted.
 - b AS 1432 for copper piping, Grade 'B' as a minimum.
 - c AS/NZS 2280 for ductile iron piping.
 - d AS/NZS 1477 for UPVC pipes.
- 2 Comply with pipe fitting manufacturing Codes:
 - a BS1640 and BS 1740 BS EN 10241
 - b ASME B16.9
 - c ASME B16.28
- 3 Comply with AS/NZS 4791/4792 for hot dipped galvanising.
- 4 Comply with the Drawings showing the indicative sizes of piping and the manner in which the system is to be installed.
- 5 Comply with the nominated system requirements:

PIPEWORK LOCATION	PIPING MATERIAL	FITTING TYPE	DESIGN WORKING PRESSURE KPA
Underground	PE100 PN16	Electrofusion	Refer Design Criteria for specific Working Pressures
	Ductile iron	Ductile iron Rubber ring joint	

Above ground exposed	Galvanised black mild steel	Malleable or roll grove couplings	
	Hot dipped galvanised after manufacture black mild steel	Welded	
Above ground internal	Black mild steel	Malleable, welded or roll groove couplings	

- 6 Not utilise on site welding.
- 7 Use long radius bends unless restricted by space. Mitre joints not permitted.
- 8 Install correctly aligned and uniformly graded.
- 9 Provide free from loose scale, burrs and swarf after fabrication.
- 10 Suitably clean prior installation, with open ends temporarily sealed with proprietary covers of pressed steel or plastic.

7.1.4 PIPING AND FITTINGS – HOSE REELS

- 1 Comply with piping manufacturing Codes:
 - a AS 1074 for steel piping, medium weight grade as a minimum. Light wall tubing not permitted.
 - b AS 1432-2004 for copper piping, Grade 'B' as a minimum.
- 2 Comply with pipe fitting manufacturing Codes:
 - a BS1640 and BS1740 BS EN 10241.
 - b ASME B16.9
 - c ASME B16.28
- 3 Comply with the Drawings showing the indicative sizes of piping and the manner in which the system is to be installed.
- 4 Comply with the nominated system requirements:

PIPEWORK LOCATION	PIPING MATERIAL	FITTING TYPE	DESIGN WORKING PRESSURE KPA
Above ground exposed	Galvanised black mild steel	Malleable or roll grove couplings	
	Hot dipped galvanised after manufacture black mild steel	Welded	
Above ground internal			

7.1.5 PIPEWORK INSTALLATION – ABOVE GROUND

- 1 Independently support for fire system exclusive use.
- 2 Not use dissimilar metals.
- 3 Use corrosion resistant supports for the environment where located.
- 4 Eliminate vibration from structure to piping and vice versa.
- 5 Standardise jointing methods for all pipework.

- 6 Install expansion joints using flexible rolled groove type fittings providing movement in all directions, located at each building expansion joint.
- 7 Provide support within 1000mm either side of a rolled groove type fitting.
- 8 Not use explosive type fixings.

7.1.6 PIPEWORK INSTALLATION – IN GROUND

- 1 Lay piping in accordance with manufacturers' instructions.
- 2 Lay piping to generally follow final ground surface contours but not to vary within:
 - a A tolerance of 20mm in a vertical plane over a distance of 6m.
 - b A tolerance of 50mm in a horizontal straight alignment within a distance of 12m.
- 3 Bed all pipework on a minimum of 150mm of approved coarse river sand.
- 4 Use long radius bends unless space restriction precludes.
- 5 Protect copper pipework and cast-iron fittings with waterproof protective pipe plastic sleeving or petrolatum wrapping in accordance with AS 2832.1:
 - a Install in strict accordance with the manufacturers' requirements.
 - b Extend protective sleeving/wrapping 200mm above the finished ground level for any piping protrusion.
 - c Equal to MAC-WRAP for protective sleeving.
 - d Equal to Denso for protective wrapping.
 - e Fit plastic sleeving to effect a smooth closely contoured envelope with sufficient slack to follow the profile of the spigot/socket interface, incorporating adequate sealing utilising PVC tape.
 - f Fit two applications of protective wrapping utilising tape widths of minimum 100mm after initial paste preparation to the pipe surface; utilise a spiral formation with a minimum 50% overlap on the first application; a reverse spiral formation over the first layer for the second application; and protective plastic sleeving as a final overlay.
 - g Install cathodic protection utilising sacrificial anodes or impressed current with facility for periodic maintenance and testing.
- 6 Temporarily cap off pipework at the end of each day's work.
- 7 Standardise jointing methods for all pipework.
- 8 Install approved thrust blocks to all UPVC type ring jointed and ductile iron cement lined piping systems with due allowance for:
 - a Approval of shop drawings indicating location and size of all thrust/anchor blocks with calculations in regard to piping size, water pressure and soil bearing pressure.
 - b Preparation of the works programme to include the required time period necessary before subjecting thrust/anchor blocks to water pressure loading.
- 9 Identify all buried pipework.
 - a Use 150mm wide poly-vinyl chloride utility service warning tape placed 150mm above the top of the service.
 - b Provide a continuous length metallic coil, colour coded to international standards with identification of the service, e.g. RED – fire hydrant service.

7.1.7 PIPING VALVES, STRAINERS AND GAUGES

- 1 Isolating Valves – In Ground:
 - a Protect with petrolatum tape wrapping or with approved manufacturers coating.
 - b Provide for remote extended key operation and operate to open position by anticlockwise movement of the spindle.
 - c Install concrete valve chamber.
 - d Access via concrete collar (top stone) surround at ground level with cast iron access lid.
 - e Identify by coloured plastic reflector fixed to cast iron lid.
- 2 Isolating Valves – Above Ground:
 - a Outside screw and yoke (OS&Y) type where required by Code.
 - b Incorporate indicator (where not OS&Y type) to identify open or closed status on sizes 100mm and greater.
 - c Incorporate visible coloured open or closed position flag indicator for butterfly type.
 - d Monitor where identified on Drawings.
- 3 Non-Return (Check) Valves:
 - a Hinged rubber faced clapper.
 - b Allow installation in horizontal or vertical position.
 - c Protect with petrolatum tape wrapping or with approved manufacturers coating where installed in ground.
- 4 Strainers:
 - a Y type.
 - b Screwed bronze body on all lines up to and including 50mm and flanged cast iron body above 50mm.
 - c Easily remove cable perforated stainless steel or brass strainer basket.
 - d Select mesh for 15kPa maximum pressure drop.
 - e Fit ball valve and drain line to waste for strainer over 50mm.
 - f Fit isolation valve and drain line to waste for strainer over 50mm.
 - g SPIRAX or approved equal.
- 5 Pressure Gauges:
 - a Install to monitor pressure at pump inlet, pump outlet, town's main inlet, fire authority booster and elsewhere as nominated.
 - b Face diameter of 100mm minimum.
 - c Display pressure in 'kPa' at 50kPa intervals.
 - d Fit with a valve cock to piping inlet.
 - e Identify with permanent label displaying gauge function.

7.1.8 FIRE HOSE REEL CABINET

- 1 Size cabinet to contain all nominated equipment (hose reel, portable fire extinguisher and backflow device as appropriate) including all clearances to comply with AS 2441.

- 2 Construct of 1.6mm thick sheet metal complete with additional stiffeners and supports to allow for fixing and rigidity.
 - 3 Provide clean, neat, non-sharp sheet metal edges free from burrs and indentations.
 - 4 Fit access door with continuous piano type hinge suitable for 180o opening.
 - 5 Secure door with non-lockable 'T' bar style chromium plated handle.
 - 6 Paint in standard manufacturers baked enamel finish or powder coated to an approved nominated colour.
 - 7 Provide signage to required size in an approved colour contrasting to cabinet background colour.
 - 8 Supply and erect cabinet support mounting column:
 - a Construct utilising mild steel minimum 75 CHS or 75 SHS.
 - b Hot dip galvanise after manufacture where exposed to weather or humid environment.
 - c Paint with primer, undercoat and finish coat to approved colour.
 - d Support within a mass concrete footing to depth of 600mm, or
 - e Support by mild steel footing plate of 300 x 300 x 10mm thick, dyna-bolted to concrete slab.
-

7.2 FIRE DETECTION SYSTEM

7.2.1 *FIRE DETECTION CONTROL AND INDICATING EQUIPMENT (FDCIE) ADDRESSABLE MICROPROCESSOR TYPE*

- 1 Addressable Microprocessor Type:
 - a Analogue addressable type, microprocessor based.
 - b Program via PC and not Eprom or keypad.
 - c Control by keypad or discreet switch type.
 - d Manual initiation of an automatic test procedure excluding ancillary equipment activation.
 - e Record time, date and events through an integral 24 columns minimum system printer.
 - f Alarm or fault condition:
 - i Write the details of the actuation to a system log file.
 - ii Print the details of the actuation to a system logging printer.
 - g Power requirements:
 - i Primary 240 V a.c.
 - ii Secondary 24 V d.c. via battery and charger.
 - iii Rated for the control and indicating equipment (CIE) loads imposed by the project.
 - iv Capacity of battery suitable for maintaining project system requirements in quiescent condition.
 - v Include ancillary loads to maximum Code requirement.
 - h Operable functions:
 - i Monitor via equipment for open circuit and closed circuit.
 - ii Provide integration for both fail safe and non-fail-safe functions in one multi-core cable with fire resistant rating.

- iii Include loop (redundant path) circuit.
- iv Include a minimum of two (2) loop circuits.
- v Provide dedicated loop circuit for Aldi and dedicated loop circuit for Woolworths.
- vi Provide a mixing of alarm/control points from various compartments/zones throughout the building, interconnected across 2-3 loops for additional redundancy.
- vii Not include all alarm/control points for a building compartment/zone connected to one (1) loop.
- viii Incorporate 25% spare capacity points per circuit loop and hardware provision, for future use.
- i Selective flexibility of input/output functions based on AND-ing or, OR-ing, NOT-ing and timing incorporating special coded operations in the resident system software programming.
- j Capable of recalling alarm and fault conditions in chronological order for the purpose of recreating an event in history.
- k Operate as standalone equipment with component communicating via an SSL approved fire-resistant peer-to-peer token ring network.
- l Connect directly to the Fire Authority or approved monitoring station, via existing ASE in building ROK06 connected to the Telstra/Optus mobile networks.
- m Enclose in cabinet with access door incorporating integral switch to isolate bell and remote equipment, excluding fire mode building plant operations.
- n Locate as shown on the Drawings.
- o Install with adequate capacity and programming sufficient to meet the requirements of the project system operations.

7.2.2 PROGRAMMING

- 1 Programming shall not require the services of a skilled software programmer but shall be loaded via a front-end interface using Basic English language instructions.
- 2 Licenses for all software shall be copied onto a USB FLASH DRIVE or similar approved device and in the event of memory loss have the ability to load back into the system via the front-end device.
- 3 Programs shall also be printed out and stored at the building with any changes to the program carried out to a strict protocol as follows:
 - a Draft changes and applicable ramifications written out in plain English, i.e. what effect does it have on the operations of the various building services approved.
 - b Changes done on site and a new USB FLASH DRIVE and print out suitably noted, kept on site.
 - c Confirmation via on-board program facility identifying changes carried out to match the written draft.
 - d Changes which are to be able to be downloaded via the main FDCIE without need to be replicated/duplicated for each other FDCIE or colour graphics system.
 - e Independent changes to the supplier's system should be identified with the tender.
- 4 An audit of the system carried out three (3) months after the commencement of the Defects Liability Period. The trends of individual devices shall be reviewed and the parameters of the device/s set to eliminate spurious alarms (including the aspirated smoke detection systems).
- 5 Devices being set to the maximum sensitivity will not be accepted unless substantiated that this setting is warranted.
- 6 Reports shall be submitted to the client representative providing details commissioning and as a result of the audit.

7.2.3 MICROPROCESSOR BASED SYSTEMS HARDWARE

- 1 Data and time notation in accordance with International Standard ISO 8601:2001.
- 2 No value for current data and/or time shall cause any interruption in system operation or functionality.
- 3 Data and time-based functionality shall behave consistently for all times.
- 4 The century in any date shall be specified either explicitly or by unambiguous algorithms or inferencing rules in all interfaces and data storage.
- 5 Leap years shall be recognised.
- 6 Ensure performance nor functionality is affected by any dates.

7.2.4 SOFTWARE

- 1 The Services Works Package shall demonstrate satisfactory operation of computerised hardware, and all software programmes intended for the project, in his/her local premises, before permission is given for loading software on site.
- 2 Programming shall not require the services of a skilled software programmer but shall be able to be loaded via a front-end interface using English language instructions.
- 3 The entire programme shall be copied into a floppy disk unit or similar approved device and in the event of memory loss, have the ability to load back into the system via the front-end device.
- 4 Hard copies shall be provided (printouts or other written records) of configuration data.
- 5 The following shall be provided in respect of ownership, support and general use of application software:
 - a Transfer copyright ownership of application software including control programs and interface drivers developed as part of these works.
 - b Unrestricted access by the building users, both on and off site, to carry out modifications and adjustment to programs, as necessary.
 - c Contract price deemed to include all royalties and ownership payable for software.
 - d Agreements of any kind not to be instituted which prevent the building users from modifying programs themselves after final completion.
 - e Latest software versions to include third party software as part of any administration package at the conclusion of Practical Completion and at any time during the warranty period, plus 12 months.

7.2.5 ZONE BLOCK PLAN

- 1 Provide adjacent the Fire Detection Control and Indicating Equipment.
- 2 Submit drawing of proposed layout for approval.
- 3 Size as required to clearly show the required information but to a minimum 500mm x 500mm.
- 4 Provide fascia finish of engraved plastic/stainless steel with all required building outlines, floor layouts and information detailed in contrasting colours.
- 5 Engrave fascia to match correct orientation layout of the building relevant to its final site location and complete with 'YOU ARE HERE' identification.
- 6 Location diagram information:
 - a Building outline for each level.
 - b Fire hydrant pumps.

- c FDCIE, and EWCIE.
- d Fire control centre.
- e Detection zones.
- f Evacuation zones.
- g Hydrants.
- h Hydrant isolation valves.
- i Mechanical and electrical switchboards.
- j Access stairs.
- k Access paths to major equipment.
- l Smoke and fire walls.
- m Significant hazards, e.g. sub stations, gas installation, flammable liquids etc.
- n The year of original installation and latest revision date.
- o A notice saying “In the event of a fire ring 000 to ensure fire service response” except the lettering height shall be not less than 5mm.
- p Statement identifying the design criteria including NCC references, Australian Standards and performance solutions.

7.2.6 DETECTORS

- 1 Initiate smoke control systems with detector selection and location to comply with AS1670.1.
- 2 Utilise 2-piece, plug into a fixed base.
- 3 Fit with an indicator to identify activation.
- 4 Provide where installed in concealed location with a remote indicator comprising a light emitting diode mounted on an engraved plastic plate identifying the location of the detector.
- 5 Protect all smoke detectors installed in air conditioning plant from the build-up of dust and contamination due to building work.
- 6 Provide for approval, a schedule of sensitivity settings where detectors have adjustable sensitivities.
- 7 Detector guards:
 - a Provide to prevent tampering and malicious damage to all detectors located in areas which are accessible to damage.
 - b Manufacture from perforated 1.6mm stainless steel sheet metal folded to form a cover over the detector.
 - c Utilise tamper resistant fixings.

7.2.7 ASPIRATED SMOKE DETECTION (ASD)

- 1 House the detector, filter and aspirating fan in an enclosure.
- 2 Locate in a cabinet specifically designed to house the detector controller as nominated on the drawings.
- 3 Include the following facilities:
 - a Minimum of two (2) alarm levels.
 - b Adjustable time delay for each alarm threshold.

- c Monitoring for detector fault, air flow fault and associated delay facilities.
 - d Relay interface cards incorporating a relay for each alarm setting and relays for detector fault, air flow fault and isolate/test mode.
- 4 Equip with a facility to enable each zone to be monitored with a standard chart recorder.
- 5 Mount the detector assembly horizontally in the fire risk area in a position to facilitate access and servicing.
- 6 Required alarm levels:
 - a Alarm Level 1 (Alert/Action). Activate a local alarm in the security office. Activate a zone fire alarm in the Fire Detection Control and Indicating Equipment and activate all warning systems.
 - b Alarm Level 2 (Fire). Call the fire brigade.
- 7 Incorporate project design requirements:
 - a Alarm levels.
 - b Initial settings for alarm levels.
 - c Obscuration range.
 - d Initial settings for alarm delay threshold.
- 8 Required initial settings for alarm levels:
 - a Alarm Level 1. 70% of bar graph reading.
 - b Alarm Level 2. 100% of bar graph reading.
- 9 Obscuration range:
 - a 0-0.2% per metre.
- 10 Required initial settings for alarm delay threshold (accumulative):
 - a Alarm Level 1. 45 seconds.
 - b Alarm Level 2. 15 seconds.
- 11 Connect the Detector Fault and Air Flow Fault alarms into the appropriate alarm zone on the Fire Detection Control and Indicating Equipment to register a fault condition on the controller and a common fault alarm on the Fire Detection Control and Indicating Equipment.
- 12 Power from the Fire Detection Control and Indicating Equipment 24 V d.c. supply or remote power supply unit.
- 13 Sampling network installation:
 - a Protect the aspirated smoke detection system from the build-up of dust and contamination due to building work.
 - b Install piping in accordance with AS 1603.4 and manufacturers' requirements.
 - c Arrange sampling pipe network to provide optimum efficiency with a time for the least favourable sampling point in the system not to exceed 60 seconds.
 - d Balance the volume of air drawn from the last sampling point to be less than 60% of the volume from the first sampling points.
 - e Submit details of the pipework design showing the proposed layout supported by calculations showing response times and balance details of each sampling point.
 - f Install main sampling pipe of white 25mm nominal diameter PVC electrical conduit type to AS 2053 identified as a fire detection sampling pipe at intervals not exceeding 600mm for the entire length.

- g Make all changes of direction with long radius 'lazy' bends and tees or purpose made branching pieces are acceptable.
- h Fit the far end of each trunk or branch pipe with an end cap drilled with a hole or holes normally of 6mm diameter, or otherwise appropriately sized to achieve the performance as specified and as calculated by the system design.
- i Seal all joints airtight using solvent cement, except at entry to enclosure.
- j Support all pipework at no less than 2000mm centres.
- k Provide sampling holes of 2mm diameter separated by intervals not more than as specified to Code or the manufacturers' recommendations.
- l Return pipework layout as designed without amendment unless approved.

14 System test:

- a Commission the entire installation under witness and with supplier/agent on site during testing.
- b Record all tests and system calibrations with copies of results retained on site in the System Logbook.
- c Introduce smoke into the detector assembly to provide a Go/No-Go Test.
- d Introduce smoke to the least favourable sampling point in each sampling point in each sampling pipe for a response time not exceeding sixty (60) seconds.
- e For more than two (2) bar graph divisions illumination under normal conditions (no smoke test), connect a chart recorder to the system and monitor for two (2) weeks from date of commissioning with a copy of recordings kept on site for future reference.
- f Where chart recordings are conducted, return in two (2) weeks to check the chart recording and make appropriate adjustments to the alarm and delay thresholds.
- g Activate appropriate Fire Detection Control and Indicating Equipment zones and confirm handover of system as fully operational.
- h Provide notification of any changes made on site to suit the structure and/or other services.
- i Fill out the logbook.

7.2.8 *MANUAL CALL POINT (MCP)*

- 1 Mount 1200 above floor level.

7.2.9 *MAGNETIC DOOR HOLDERS*

- 1 Electro-magnetic operation on signal from FDCIE.
- 2 24 V d.c. power supply from FDCIE.
- 3 Manual push button release at holder location.
- 4 Automatic release on system failure or power supply failure.

7.2.10 *EXTERNAL ALARM VISUAL ALARM DEVICE*

- 1 Suitable for external environments.
- 2 Be active when the FDCIE is in the fire alarm condition.
- 3 Flash RED in colour with a sign 'FIRE in white lettering on a red background. Lettering height shall not be less than 50mm. The sign shall be weather and fade resistant.

- 4 Locate to identify the FDCIE access point.

7.2.11 ELECTRICAL STRIKE LATCHES

- 1 Mortice type.
- 2 Fail safe (power loss).
- 3 Comply with AS 1905.1 where applicable to installations on fire or smoke resistant doors.
- 4 Compatible for required specified DC voltage.
- 5 Operate directly from the FDCIE or through the security system as specified.
- 6 LOCKWOOD 3500 SERIES or equal approved.

7.2.12 SYSTEM INTERFACE CONTROL

- 1 Provide all fire detection system interface controls to specified requirements associated with building fire safety and plant operation.

7.3 PORTABLE FIRE EQUIPMENT

7.3.1 PORTABLE FIRE EXTINGUISHERS

- 1 Select type and capacity of fire extinguishers as follows/detailed on the drawings
- 2 Provide all signage and mounting brackets.

AREA	TYPE	SAA RATING	MINIMUM CAPACITY
Electrical Switch Room	CO2	10B (E)	5.0 kg
Plantroom	Dry Chemical	6A:60B (E)	4.5 kg
Kitchen	Dry Chemical Wet Chemical	6A:60B (E) 3A:4F	4.5 kg 7.0 ltr
Loading Bay	Dry Chemical	6A:60B (E)	4.5 kg
Car Park	Dry Chemical	6A:60B (E)	4.5 kg
Offices	Dry Chemical	6A:60B (E)	4.5 kg

7.3.2 FIRE BLANKETS

- 1 Provide all signage and mounting brackets.

7.4 OCCUPANT OR EMERGENCY WARNING SYSTEM

7.4.1 GENERAL REQUIREMENTS

- 1 Interface with fire alarm signal input from Fire Detection Control and Indicating Equipment.
- 2 Zone for compatibility with the detection and/or suppression system alarm zones associated with smoke or fire compartments.
- 3 Sound levels:
 - a Comply with the requirements of AS 1670.1 and BCA.

- b Transmitted to all areas of the building to achieve a minimum of 10dB above the noisiest background sound pressure level. Note:

A sound level of 105dBA is the maximum under AS 1670.1. Some projects which have high ambient noise environments may need up to 120dBA and this shall be specifically noted for the project.

4 Sound tones shall:

- a Generate an evacuation signal complying with the AS 1670 specification for automatic emission frequency and cycle.
- b Include an approved customisable voice message to suit the project requirements complying with the specification requirements of AS 1670 Submit proposed voice message for approval.
- c Include facility for local manual microphone (public address) intervention.

7.4.2 *EMERGENCY WARNING CONTROL AND INDICATING EQUIPMENT (EWCIE)*

1 Microprocessor based:

- a Utilise discrete switch type control.
- b Utilise keypad type control complying with requirements for Fire Authority application.

2 Locate in Fire Control Centre.

3 Operate from a 240V power supply.

4 Include emergency public address facility.

5 Include one amplifier per evacuation zone.

6 Include one tone generator per evacuation zone.

7 Include a voice message for both evacuation and alert signals capable of being customised to suit the project requirements. Submit proposed voice messages for approval.

8 Include adjustable time delay from 'alert' to 'evacuation'.

9 Include spare capacity of 25% on each zone amplifier.

10 Include spare capacity of 25% on each tone generator.

11 Incorporate capacity for number of evacuation zones to meet system requirements plus additional 25% zones allowance.

12 Include hardware fitted within the panel to comply with specified spare capacity requirements.

13 Include one spare amplifier and tone generator within the panel. Edit as required depending on the future expansion for the project.

14 Isolation key switches compatible with FDCIE key switches.

15 Display emergency notice on instructions for operation.

7.4.3 *EMERGENCY INTERCOM CONTROL AND INDICATING EQUIPMENT (EICIE)*

1 Microprocessor based:

- a Utilise discrete switch type control.
- b Utilise keypad type control complying with requirements for Fire Authority application.

2 Locate in Fire Control Centre.

3 Operate from a 240V power supply.

- 4 Include emergency public address facility.
- 5 Include one amplifier per evacuation zone.
- 6 Include one tone generator per evacuation zone.
- 7 Include a voice message for both evacuation and alert signals capable of being customised to suit the project requirements. Submit proposed voice messages for approval.
- 8 Include adjustable time delay from 'alert' to 'evacuation'.
- 9 Include spare capacity of 25% on each zone amplifier.
- 10 Include spare capacity of 25% on each tone generator.
- 11 Incorporate capacity for number of evacuation zones to meet system requirements plus additional 25% zones allowance.
- 12 Include hardware fitted within the panel to comply with specified spare capacity requirements.
- 13 Include one spare amplifier and tone generator within the panel. Edit as required depending on the future expansion for the project.
- 14 Isolation key switches compatible with FDCIE key switches.
- 15 Display emergency notice on instructions for operation.

7.4.4 OCCUPANT WARNING SPEAKERS

- 1 Recessed cone speakers installed in all ceiling areas.
- 2 Cone speakers of 100mm dia. Installed recessed or within surface mounting fixture.
- 3 Horn type speakers installed in plantrooms and similar non habitable service type areas.
- 4 Clearly marked to distinguish from other types of speakers, e.g. FIRE or PA/FIRE.
- 5 Incorporate multiple tapping facility.
- 6 Incorporate fault monitoring to circuits.

7.4.5 VISUAL ALARM DEVICES

- 1 Install in areas dedicated to students and teaching areas.
- 2 Installed within high ambient noise areas with a background noise level exceeding 85dBA.
- 3 Utilise AS7240.23 compliant visual alarm device (VAD) and install in accordance with the area of coverage rating.
- 4 Incorporate fault monitoring to circuits.
- 5 Install in areas designated for hearing impaired persons.
- 6 Install in areas where hearing protection devices are required to be worn.

7.4.6 ELECTRICAL

- 1 Comply with Electrical requirements of the Fire Detection and Alarm Section of this specification.
- 2 Utilise fire resistant cabling where passing through any evacuation zone, en-route to each final evacuation zone circuit destination.
- 3 Utilise non fire-resistant cabling only where contained within the evacuation zone of the final circuit destination.

7.5 ELECTRICAL SYSTEMS

7.5.1 GENERAL REQUIREMENTS

- 1 Provide electrical services associated with power, control and alarm systems for fire services installations.
- 2 Deliver on site, where possible, in sealed unopened coils or drums with makers' seals and labels intact.
- 3 Select cables with full consistent colour code and of appropriate grade for the service.
- 4 Use cabling:
 - a Minimum 1.5mm² TPS.
 - b Minimum 1.5mm² fire resistant.
 - c Red sheathed for fire detection.
 - d Screened RS485 data loop for communication.
 - e Double insulated 'figure 8' for occupant warning.
- 5 Provide a complete cable support system comprising trays/conduit/ducting including brackets, fixings and accessories.
- 6 Fabricate brackets, racks and hangers using structural steel sections or other materials in sections of equivalent strength.
- 7 Ensure a rigid support system for trays/conduit/ducting.
- 8 Provide metallic conduit installations within all plantrooms where located below 2m above floor level.
- 9 Support wiring in vertical runs for any system type using retaining clips at intervals of not more than 1m.
- 10 Support any system type at maximum interval of 1.5m or in accordance with manufacturers' recommendations.
- 11 Not use explosive fixings.
- 12 Run cables in a manner eliminating any possibility of strain on the cable.
- 13 Use dedicated penetrations for wiring run through building elements.
- 14 Test all cables after installation.

7.5.2 INSTALLATION METHODS

LOCATION/WALL CONSTRUCTION	INSTALLATION AND CONCEALED CABLING FACILITIES
Concrete block work	Conduit in cavity.
Concrete floor slabs without ceiling space	Conduit embedded in slab.
Face brick external cavity wall	Conduit run in cavity and tied against inner brick surface.
Stud partition	Wiring run through cavity, not in conduit.
Suspended ceiling/roof spaces	Wiring fixed to structure over, supported at intervals of not more than 1.5m and not to lay on the top side of suspended ceilings or light fittings. Multiple cables neatly grouped together and run in the same direction. More than 5 cables run on a cable tray together.
Vertical riser	Wiring run on cable tray or in cable duct.

LOCATION/WALL CONSTRUCTION	INSTALLATION AND CONCEALED CABLING FACILITIES
Plant and services areas	Wiring may be run on surface but concealed.
Accessible roof spaces	Wiring run in conduit.
Between structure and equipment	Wiring run in flexible conduit.

7.5.3 *INSTALLATION – DUCTED ENCLOSURE SYSTEMS*

- 1 Provide galvanised, purpose made and rigidly supported.
- 2 Incorporate segregation where used for multiple services.
- 3 Provide PVC bushes for metallic cable entries.
- 4 Provide purpose made accessories to match the duct system using screw fixed or clip on covers removable only with the use of tools.

7.5.4 *INSTALLATION – CABLE TRAY SYSTEMS*

- 1 Galvanised steel type.
- 2 Minimum steel thickness:
 - a Trays <150mm wide: 1mm
 - b Trays ≥150mm, <300mm wide: 1.2mm
 - c Trays ≥300mm wide: 1.6mm
- 3 Admiralty pattern, reverse stamping perforations.
- 4 Provide purpose made proprietary bends and tees.
- 5 Fix supports to the building structure or fabric using direct fixing hangers or brackets.
- 6 Provide a minimum of 150mm free space above and 600mm free space on one side of the trays.
- 7 Provide slats or rails suitable for fixing cable ties, strapping or saddles.
- 8 Provide bends with a minimum inside radius of twelve (12) times the outside diameter of the largest cable diameter carried.
- 9 Provide rounded non-sharp edge support surfaces under cables at exit and entry.
- 10 Use stainless steel straps on MIMS and other fire rated cables.

7.5.5 *INSTALLATION – PVC CONDUITS*

- 1 Minimum size of 20mm.
- 2 Provide straight long runs, smooth and free from rags, burrs and sharp edges.
- 3 Set conduits to minimise the number of fittings.
- 4 Install conduits in parallel runs with right angle changes of direction where exposed to view.
- 5 Locate inspection fittings in accessible positions.
- 6 Provide draw cords in conduits not used and leave 1m of coiled cord secured at each end of the run.
- 7 Provide draw in boxes at intervals not exceeding 30m in straight runs and at changes of level of direction.
- 8 Seal penetrations around conduits and around cables within sleeves.

- 9 Use proprietary two (2) hole type saddles at 1.2m centres for surface run conduit.
- 10 In concrete slabs:
 - a Not run in concrete toppings or within pre tensioning cable zones.
 - b Cross pre tensioning cable zones at right angles.
 - c Avoid crossovers and minimise the number in any location with parallel conduits spaced at least 50mm apart.
 - d Minimum concrete cover of 20mm or equivalent to the conduit diameter.
 - e Fix directly to top of the bottom layer of reinforcing where passing above a single layer of reinforcing.
 - f Not run in the floor slabs of boiler rooms, plant rooms and tank rooms.
 - g Locate in the core filled sections of precast hollow block type floors.
- 11 Within concrete columns:
 - a Not more than four (4) 25mm maximum diameter conduits located centrally.
 - b Use bends with minimum radius of 150mm on entry.
 - c Not utilise chases.

7.5.6 INSTALLATION – METALLIC CONDUITS

- 1 Paint ends and joint threads with zinc rich organic binder to GPC-C-29/16.
- 2 Provide flexible couplings comprising flexible conduit and fittings at structural expansion joints and in long straight runs where ambient temperature varies by more than 40°C.
- 3 Maintain electrical conductivity between the two (2) ends of rigid metallic conduit.
- 4 Not affect any equipment and/or appliances provided under these works through interference caused by radio, mobile phones or other electronic transmitting or receiving equipment.

7.5.7 INSTALLATION – FIRE RESISTANT SYSTEMS

- 1 Install in accordance with manufacturers' recommendations.
- 2 Install such that mechanical damage is unlikely to occur or provide protection to AS 3013.
- 3 Fix securely with stainless steel clips to trays or fire-resistant building elements.
- 4 Provide metal glands or close-fitting metallic bushes for cable entry to junction boxes and switchboards.
- 5 Meet or exceed WS52W installation category.

7.5.8 INSTALLATION – CABLE IN TRENCHES

- 1 Comply with excavation, bedding and backfilling specified requirements.
- 2 In-ground conduits:
 - a Seal buried entries to conduits using waterproof seals and/or conduit fittings.
 - b Seal spare conduits immediately after installation and seal other ducts and conduits after cable installation.
- 3 Cable pits:
 - a Provide proprietary concrete or polymer moulded pits for size $\leq 1.2 \times 1.2\text{m}$.
 - b Construct in situ walls and bottom using core filled reinforced block work or 75mm thick reinforced concrete incorporating waterproofing agent in render or concrete for size $> 1.2 \times 1.2\text{m}$.

4 Pit covers:

- a** Provide to suit expected loads and fit flush with the top of the pit and finished ground level.
- b** Maximum weight not exceeding 40kg for any section of the cover.
- c** Provide a lifting handle for each cover size.

5 Drainage of pits:

- a** Provide adequate drainage from the bottom of cable pits, either to absorption trenches filled with rubble or to the stormwater drainage system.
- b** Construct minimum size 300 x 300 x 2000mm absorption trenches.

6 Route marking:

- a** Accurately record routes of underground cables before backfilling.
- b** Accurately mark the location of underground cables using route markers comprising marker plates set flush in a concrete base.
- c** Place markers at each joint, route junction, change of direction, termination and building entry point and in straight runs at intervals of not more than 100m.
- d** Set markers in minimum size 200mm diameter/square x 200mm deep concrete.
- e** Show direction of cable run using direction arrows on the marker plate.
- f** Indicate distance to the next marker on the marker plate.
- g** Use brass plates minimum size 75 x 75 x 2mm thick fixed in position with waterproof adhesive and four (4) brass or stainless-steel countersunk screws.
- h** Set marker plate flush with paved surfaces and 25mm above other surfaces.